A Mathematical Finance Field Guide

From Borel Sets and Itô Calculus to Local Volatility, Path Signatures, and Identifiability

A structured, experiment-driven introduction to the mathematical objects, financial contracts, computational methods, and inverse-problem ideas that connect modern quantitative finance.

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This text is educational and is not investment, trading, legal, or accounting advice. Model prices depend on assumptions and calibration choices.

How to use this guide

The terms in the title are not a random vocabulary list. They form a chain:

$$\text{measurable events} \longrightarrow \text{random paths} \longrightarrow \text{stochastic dynamics}$$
$$\longrightarrow \text{contract payoff} \longrightarrow \text{price map} \longrightarrow \text{calibration and risk.}$$

A Borel set makes events measurable. A stochastic process produces paths. An SDE specifies their local dynamics. A derivative converts a path into a payoff. Risk-neutral valuation converts that payoff into a price. A calibration algorithm then tries to invert the price map and recover parameters such as an implied-volatility surface or a local-volatility function.

This guide is designed for three passes:

1. **First pass: intuition.** Read the boxed definitions, figures, and chapter summaries. Skip derivations on first contact.
2. **Second pass: mathematics.** Work through the equations and check dimensions, conditioning information, and assumptions.
3. **Third pass: computation.** Reproduce the experiments, change one assumption at a time, and inspect which conclusions survive.

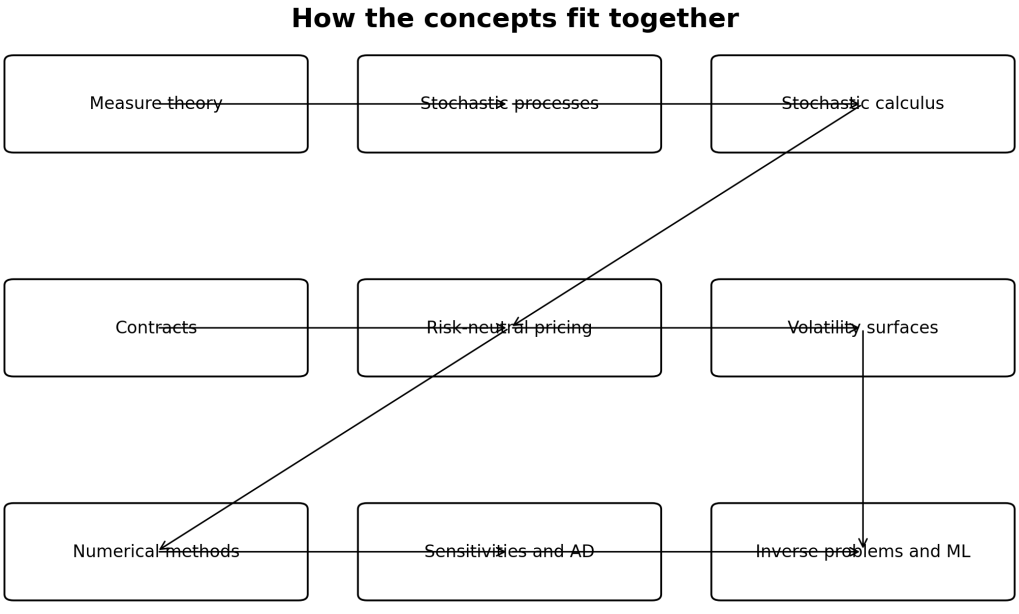


Figure 1: The main conceptual dependencies in the guide.

A compact learning map

Stage	Central question	Main objects
Foundations	What counts as an observable event or integrable quantity?	Borel sets, sigma-algebras, measures, kernels, complete spaces
Dynamics	How does uncertainty evolve through time?	filtrations, paths, Brownian motion, martingales, SDEs, particles
Contracts	How does a financial rule transform a path into cash flows?	calls, puts, strikes, moneyness, American and exotic options
Pricing	How does no-arbitrage determine value?	risk-neutral measures, Black-Scholes, martingale representation, Monte Carlo
Volatility	What does "volatility" mean in different models?	realized, implied, local, stochastic volatility; Dupire equation
Computation	How are prices and sensitivities obtained numerically?	computational graphs, Greeks, LSMC, splines, convolutions
Inference	Can model inputs be recovered from observed prices?	price maps, Jacobians, rank, local injectivity, manifolds, gauge freedom
Approximation	How can functions of states or paths be represented?	polynomials, splines, kernels, MLPs, signatures, teacher-student models

Conventions

Unless stated otherwise, time is measured in years; rates and volatilities are continuously compounded and annualized; S_t denotes an underlying price; K is a strike; T is maturity; r is the continuously compounded risk-free rate; q is a continuous dividend yield; and W_t is a standard Brownian motion. The notation $\mathbb{P}$ is used for a physical probability measure and $\mathbb{Q}$ for a risk-neutral measure.

What this guide means by a theorem

A theorem is never just a formula. Its assumptions are part of its content. Optional stopping can fail without integrability or boundedness conditions. An inverse function theorem gives a local conclusion, not necessarily a global one. A universal approximation theorem guarantees representability, not easy training, data efficiency, stability, or extrapolation.

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Notation at a glance

Symbol	Meaning
Ω	sample space: the set of possible elementary outcomes
$\mathcal{F}$	sigma-algebra of measurable events
$\mathcal{B}(\mathbb{R}^d)$	Borel sigma-algebra generated by open subsets of $\mathbb{R}^d$
$\mathbb{P}, \mathbb{Q}$	probability measures; usually physical and risk-neutral respectively
$\mathbb{E}^{\mathbb{P}}, \mathbb{E}^{\mathbb{Q}}$	expectations under the indicated measure
$\{\mathcal{F}_t\}_{t\geq 0}$	filtration: information available through time t
W_t	standard Brownian motion
X_t	generic stochastic process or state variable
S_t	underlying asset price
K, T	option strike and maturity
r, q	risk-free rate and continuous dividend yield
μ, σ	drift and volatility parameters
C, P	call and put values
$F_{t,T}$	forward price for delivery at T
Δ, Γ, ν	Delta, Gamma, and Vega sensitivities
$J(\theta)$	Jacobian of a vector-valued map at parameter θ
$\ker A$	nullspace of a linear map A
$\operatorname{rank} A$	dimension of the image of A
μ_N	empirical measure of N particles
$\mathcal{S}(X)$	signature series of a path X

Three meanings of "complete" that must not be confused

A *complete measure space* contains every subset of every null set. A *complete metric space* contains the limit of each of its Cauchy sequences. A *complete market* allows every admissible contingent claim to be replicated. These ideas are related only through context, not through a single definition.

Chapter 1

Measurable foundations: Borel sets, measures, kernels, and complete spaces

1.1 Why finance begins with sets

A mathematical model starts with a set Ω of possible outcomes. In a one-period coin model, an outcome might be heads or tails. In a continuous-time market, an outcome is better imagined as an entire possible history of prices, rates, and other state variables. A derivative payoff is then a function of that history.

The first difficulty is that one cannot assign probabilities consistently to every conceivable subset of a rich sample space while retaining the properties needed by calculus. Measure theory therefore separates all subsets from the subsets that are declared *measurable*.

Sigma-algebra

A collection $\mathcal{F}$ of subsets of Ω is a *sigma-algebra* if:

1. $\Omega \in \mathcal{F}$;
2. $A \in \mathcal{F}$ implies $A^c \in \mathcal{F}$;
3. $A_1, A_2, \ldots \in \mathcal{F}$ implies $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

By De Morgan's laws, $\mathcal{F}$ is also closed under countable intersections. Elements of $\mathcal{F}$ are called measurable sets or events.

The word "countable" is essential. Probability theory repeatedly takes limits of sequences: increasingly fine time grids, truncations of unbounded payoffs, or approximations of stopping times. Closure under countable operations makes these limits compatible with measurable events.

1.2 Borel sets

Suppose the state space has a topology, such as the usual topology on $\mathbb{R}^d$. Open sets express geometric nearness. The smallest sigma-algebra containing all open sets is the Borel sigma-algebra.

Borel sigma-algebra and Borel set

For a topological space E, the *Borel sigma-algebra* $\mathcal{B}(E)$ is the sigma-algebra generated by the open subsets of E. A *Borel set* is any member of $\mathcal{B}(E)$.

On $\mathbb{R}$, every open interval, closed interval, singleton, countable set, closed set, and every countable union or intersection built from such sets is Borel. There also exist subsets of $\mathbb{R}$ that are not Borel.

They are too irregular to be generated from open sets using only countably many sigma-algebra operations [1, 2].

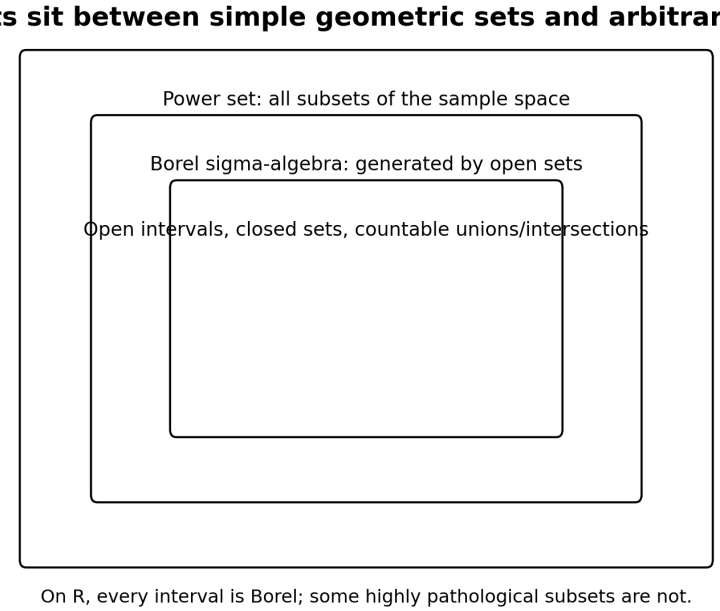


Figure 1.1: The Borel sigma-algebra contains all sets generated from ordinary geometric sets by countable set operations, but it need not contain every subset.

Why do Borel sets matter in finance? If S_t is a real-valued price, statements such as

$$\{S_T > K\}, \qquad \left\{\max_{0\le t\le T} S_t < B\right\}, \qquad \left\{\int_0^T S_t \,\mathrm{d}t > A\right\}$$

should be events to which probabilities can be assigned. This is guaranteed by constructing the process and the path functionals measurably with respect to suitable Borel sigma-algebras.

1.2.1 Borel versus Lebesgue measurable

The Borel sigma-algebra is generated by topology. The *Lebesgue sigma-algebra* on $\mathbb{R}^d$ is the completion of the Borel sigma-algebra under Lebesgue measure: it additionally contains every subset of every Borel set of Lebesgue measure zero. Thus every Borel set is Lebesgue measurable, but not every Lebesgue measurable set is Borel.

This distinction rarely changes a basic option-pricing calculation, but it matters in rigorous probability because two random variables that differ only on a null set are normally treated as equivalent.

1.3 Measures and probability measures

Measure

A measure on $(\Omega, \mathcal{F})$ is a function $\mu : \mathcal{F} \to [0, \infty]$ satisfying:

1. $\mu(\varnothing) = 0$;
2. for pairwise disjoint $A_1, A_2, \ldots \in \mathcal{F}$,

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

A probability measure is a measure with $\mu(\Omega) = 1$.

Measures generalize length, area, mass, and probability. A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$. The three components answer different questions:

- Ω: what outcomes are possible?
- $\mathcal{F}$: which questions about outcomes are measurable?
- $\mathbb{P}$: how much probability is assigned to each measurable event?

A function $X : \Omega \to \mathbb{R}$ is a random variable when it is measurable, meaning

$$X^{-1}(B) = \{\omega : X(\omega) \in B\} \in \mathcal{F} \quad \text{for every } B \in \mathcal{B}(\mathbb{R}).$$

This definition says that every Borel question about the numerical output of X corresponds to a measurable event in the original sample space.

The distribution of X is the *pushforward measure*

$$\mathbb{P}_X(B) = \mathbb{P}(X \in B) = \mathbb{P}(X^{-1}(B)).$$

One can therefore work either with the abstract outcome ω or directly with the law of X.

1.3.1 Integration and expectation

Expectation is integration with respect to probability:

$$\mathbb{E}^{\mathbb{P}}[X] = \int_{\Omega} X(\omega)\, \mathbb{P}(\,\mathrm{d}\omega).$$

For a payoff H, the risk-neutral pricing formula will later take the form

$$V_t = B_t \mathbb{E}^{\mathbb{Q}}\left[\frac{H}{B_T} \,\middle|\, \mathcal{F}_t\right],$$

where B_t is a numeraire account. This expression combines measure, conditional expectation, filtration, and discounting in a single line.

1.3.2 Conditional expectation

The conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ is not primarily a number. It is a $\mathcal{G}$-measurable random variable satisfying

$$\int_A \mathbb{E}[X \mid \mathcal{G}]\, \mathrm{d}\mathbb{P} = \int_A X\, \mathrm{d}\mathbb{P}, \qquad A \in \mathcal{G}.$$

It is the best mean-square predictor of X among $\mathcal{G}$-measurable random variables when $X \in L^2$. The tower property

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}], \qquad \mathcal{H} \subseteq \mathcal{G},$$

is the algebraic engine behind dynamic programming and martingale pricing.

1.4 Three forms of completeness

The word *complete* is overloaded. Keeping its meanings separate prevents major conceptual errors.

1.4.1 Complete measure spaces

A measure space $(\Omega, \mathcal{F}, \mu)$ is complete if, whenever $N \in \mathcal{F}$ has $\mu(N) = 0$, every subset $A \subseteq N$ also belongs to $\mathcal{F}$. Completing a measure space adds all such subsets. The completion does not change integrals of already measurable functions; it ensures that modifications on null sets remain measurable.

1.4.2 Complete metric and normed spaces

A sequence (x_n) in a metric space is Cauchy if its elements become arbitrarily close to each other. The space is complete if every Cauchy sequence converges to a point inside the space.

A complete normed vector space is a *Banach space*. A complete inner-product space is a *Hilbert space*. Important examples are:

$$L^p(\Omega, \mathcal{F}, \mathbb{P}), \quad 1 \le p \le \infty,$$

with L^2 a Hilbert space under

$$\langle X, Y \rangle = \mathbb{E}[XY].$$

Completeness matters because optimization algorithms and approximation sequences should converge to objects that still live in the model space.

1.4.3 Complete markets

A financial market is complete if every admissible contingent claim can be replicated by trading available assets. In standard Brownian models, completeness is linked to a martingale representation theorem and to uniqueness of the equivalent risk-neutral measure. This is not the same concept as completeness of L^2, even though the proof machinery uses complete function spaces.

1.5 What is a kernel? Four common meanings

Context	Definition	Why it matters
Linear algebra	$\ker A = \{x : Ax = 0\}$	Directions invisible to a linear map; central to identifiability

Probability	$K(x, A)$ is a probability measure in A for each state x and measurable in x	Describes a transition law or conditional distribution
Integral operators	$(Tf)(x) = \int k(x, y) f(y)\,\mathrm{d}y$	Represents smoothing, pricing, Green functions, and inverse problems
Machine learning	positive-definite $k(x, y) = \langle \phi(x), \phi(y) \rangle$	Defines similarities and a reproducing-kernel Hilbert space

1.5.1 Kernel as nullspace

If $A\theta = y$, then any vector $v \in \ker A$ satisfies

$$A(\theta + v) = A\theta.$$

The data cannot distinguish θ from $\theta + v$. A zero kernel is equivalent to injectivity for a linear map. In nonlinear calibration, the kernel of the Jacobian gives locally invisible parameter directions.

1.5.2 Markov kernels

A Markov kernel $K(x, A)$ assigns to a current state x a probability of moving into event A. For a Markov chain,

$$\mathbb{P}(X_{n+1} \in A \mid X_n = x) = K(x, A).$$

For diffusion processes, a transition density $p(s, x; t, y)$ is a density representation of such a kernel:

$$K_{s,t}(x, A) = \int_A p(s, x; t, y)\,\mathrm{d}y.$$

1.5.3 Integral and convolution kernels

A convolution has the form

$$(f * k)(x) = \int f(x - y) k(y)\,\mathrm{d}y.$$

The kernel k says how nearby values are weighted. A Gaussian kernel smooths; a finite-difference kernel detects slopes; a transition-density kernel propagates a distribution through time.

1.5.4 Positive-definite kernels and completeness of features

A positive-definite kernel can be represented as an inner product in a feature space. A kernel is called *universal* on a compact domain when its reproducing-kernel Hilbert space is dense in the continuous functions under the uniform norm. A related probabilistic notion is a *characteristic* kernel, for which the kernel mean embedding uniquely identifies probability measures. Informally, these properties mean the feature family is rich enough to separate functions or distributions. They do not mean that a finite noisy sample makes estimation easy [23, 25].

1.6 Dimensionless variables and dimensionless functions

A physical quantity has units: dollars, years, shares, or inverse years. A quantity is dimensionless if its units cancel. Ratios such as K/S_0, log-moneyness $\log(K/F)$, and standardized returns are dimensionless.

Dimensionless function

A function is dimensionless when its output has no physical units, usually because both its inputs and output have been normalized into unit-free combinations. Nondimensionalization rewrites a model in terms of such combinations.

Why do this?

- It prevents equations from adding incompatible units.
- It reveals scale invariances.
- It reduces the number of effective parameters.
- It improves numerical conditioning because variables lie on comparable scales.

For a Black-Scholes call, homogeneity gives

$$C(\lambda S, \lambda K, T, r, q, \sigma) = \lambda C(S, K, T, r, q, \sigma).$$

Therefore

$$\frac{C}{S} = c\left(\frac{K}{S}, T, r, q, \sigma\right).$$

The function c is dimensionless.

Experiment: Scale collapse of Black-Scholes prices

The figure computes call prices for $S_0 = 50, 100, 200$ while keeping K/S_0, maturity, rates, and volatility fixed. The normalized prices C/S_0 collapse onto the same curve. This is a numerical display of homogeneity, not a fitted coincidence.

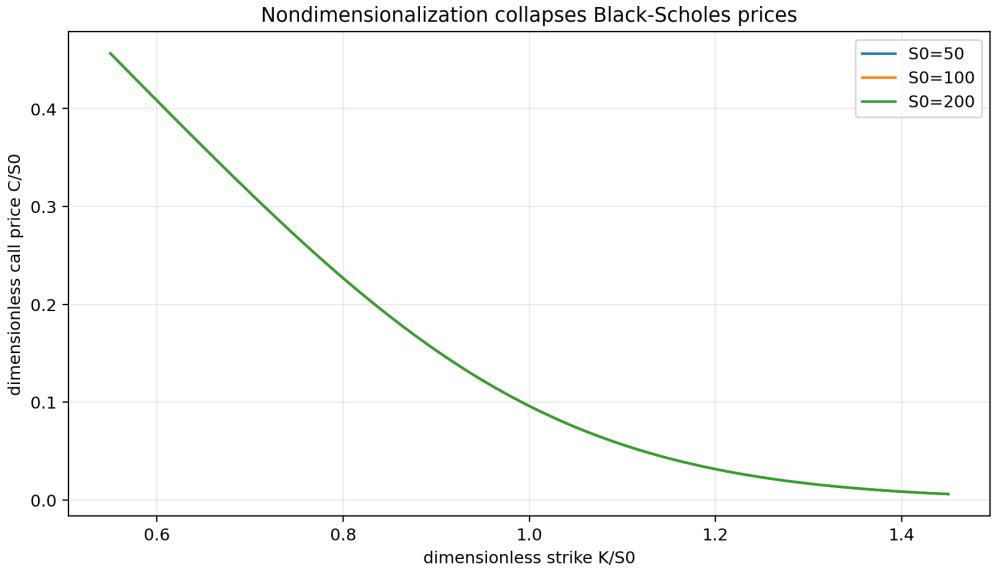


Figure 1.2: Nondimensionalization exposes the scale invariance of Black-Scholes prices.

1.7 Manifolds and maps: a first geometric view

A d-dimensional manifold is a space that looks locally like $\mathbb{R}^d$ even if its global geometry is curved. The surface of a sphere is two-dimensional because a small patch can be described by two coordinates.

In calibration, the set of model outputs may form a manifold inside a much larger data space. If a parameter $\theta \in \mathbb{R}^d$ generates prices

$$\mathcal{P}(\theta) = (P_1(\theta), \ldots, P_m(\theta)) \in \mathbb{R}^m,$$

then the image $\mathcal{P}(\Theta)$ is often a curved d-dimensional surface in $\mathbb{R}^m$ when the Jacobian has rank d. This geometric language makes ideas such as tangent directions, invisible directions, and local identifiability precise [20].

The recurring forward-map pattern

Most later concepts fit one template. A hidden object θ - a volatility, a function, or a path law - is passed through a forward map $\mathcal{P}$ to produce observable prices. Pricing studies $\theta \mapsto \mathcal{P}(\theta)$. Calibration studies the inverse direction. The kernel and singular values of the derivative of $\mathcal{P}$ tell us what information is lost.

Chapter summary

- Borel sets are the measurable sets generated by topology.
- A measure assigns consistent size; a probability measure has total mass one.
- Random variables are measurable maps, and their laws are pushforward measures.
- Complete measure spaces, complete metric spaces, and complete markets are different notions.
- "Kernel" can mean a nullspace, transition law, integral weight, or positive-definite similarity.
- Dimensionless variables expose invariances and improve numerical stability.
- A price map often traces a low-dimensional manifold in a high-dimensional observation space.

Chapter 2

Stochastic processes, paths, martingales, shocks, and particles

2.1 From random variables to stochastic processes

A random variable gives one uncertain number. A stochastic process gives an uncertain number at every time.

Stochastic process

A stochastic process on $(\Omega, \mathcal{F}, \mathbb{P})$ with state space E is a collection

$$X = \{X_t : t \in I\}$$

of E-valued random variables indexed by time or another parameter.

There are two equivalent viewpoints:

- Fix t: $X_t(\omega)$ is a random variable across outcomes.
- Fix ω: $t \mapsto X_t(\omega)$ is a *sample path* or *trajectory*.

A model specifies a probability distribution over paths. A path-dependent derivative reads the whole trajectory, not only its terminal value.

2.2 Filtrations and information

A filtration $\{\mathcal{F}_t\}_{t \geq 0}$ is an increasing family of sigma-algebras:

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}, \qquad s \leq t.$$

It represents information available through time t.

A process X_t is *adapted* if X_t is $\mathcal{F}_t$-measurable. In plain language, its current value does not require future information. An integrand in an Itô integral must be predictable or at least progressively measurable under suitable conditions; this blocks trading rules that look into the future.

A filtration is part of the model

The same numerical process can be a martingale under one filtration and not under another. Information determines what is predictable. Finance therefore models not only prices but also what agents are allowed to know when making a decision.

2.3 Brownian motion

Standard Brownian motion

A process W_t is standard Brownian motion if:

1. $W_0 = 0$ almost surely;
2. it has independent increments;
3. $W_t - W_s \sim \mathcal{N}(0, t-s)$ for $0 \le s < t$;
4. its paths are almost surely continuous.

Brownian motion is continuous but almost surely nowhere classically differentiable. Over a short interval Δt,

$$\Delta W \sim \sqrt{\Delta t}\, Z, \qquad Z \sim \mathcal{N}(0,1).$$

The random increment is of order $\sqrt{\Delta t}$ rather than Δt. Squaring produces a term of order Δt, which survives in the limit and creates the correction in Itô's lemma.

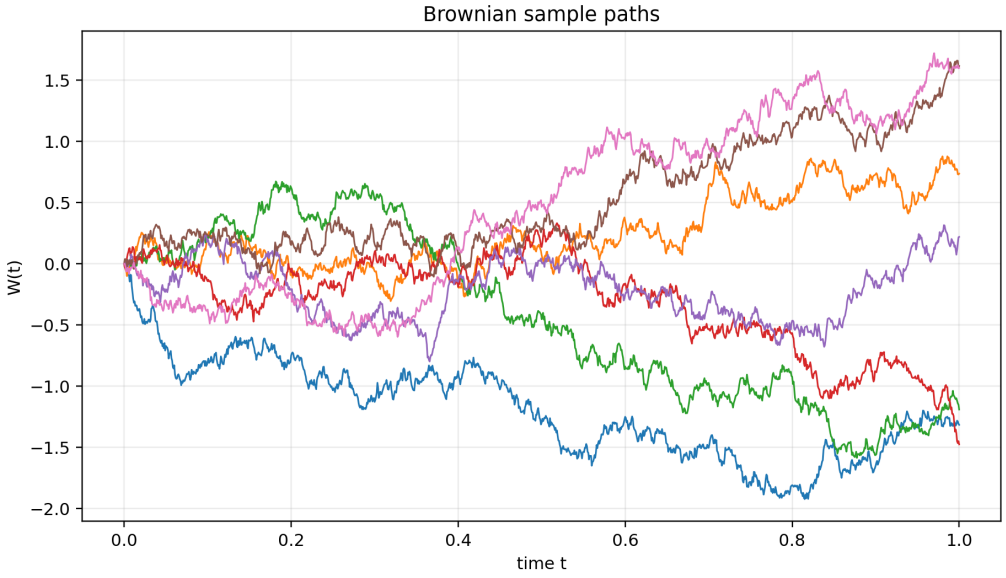


Figure 2.1: Different Brownian trajectories drawn from the same law. Continuity does not imply smoothness.

2.3.1 Quadratic variation

For a partition $0 = t_0 < \cdots < t_n = T$, define

$$[W]_T^{(n)} = \sum_{i=0}^{n-1} (W_{t_{i+1}} - W_{t_i})^2.$$

As the mesh tends to zero,

$$[W]_T^{(n)} \longrightarrow T$$

in probability, and along suitable nested partitions almost surely. In differential shorthand, $(\mathrm{d}W_t)^2 = \mathrm{d}t$. This is not ordinary algebra; it summarizes a limiting quadratic-variation rule [3, 4].

Experiment: Quadratic variation

For each partition size, 4,000 Brownian paths were simulated on $[0, 1]$. With 2,048 intervals, the sample mean of the squared-increment sum was 1.00014 with standard deviation 0.03150, close to the theoretical limit 1.

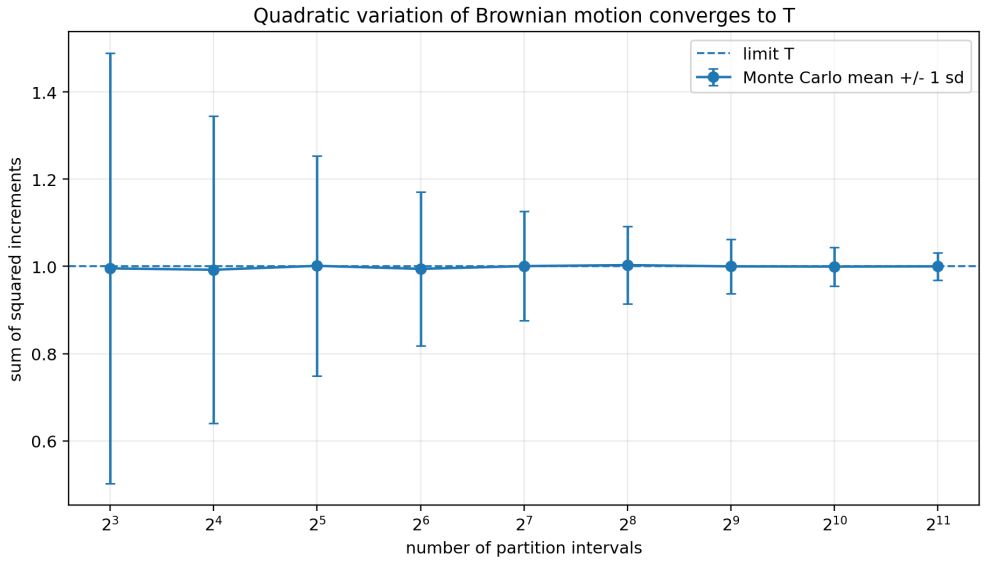


Figure 2.2: The mean quadratic variation approaches $T = 1$, while its dispersion shrinks as the partition is refined.

2.4 Martingales

Martingale

An integrable adapted process M_t is a martingale with respect to $(\mathcal{F}_t, \mathbb{P})$ if

$$\mathbb{E}^{\mathbb{P}}[M_t \mid \mathcal{F}_s] = M_s, \qquad s \le t.$$

It is a submartingale if the conditional expectation is at least M_s and a supermartingale if it is at most M_s.

A martingale has no predictable conditional drift under the specified measure and filtration. This does not mean it is constant or riskless. Brownian motion is a martingale, as is the stochastic integral $\int_0^t H_s \, \mathrm{d}W_s$ under square-integrability conditions.

2.4.1 Which "martingale theorem"?

There is no single theorem with that name. Four results are especially important.

2.4.1.1 Optional stopping theorem

Under conditions such as bounded stopping times, uniform integrability, or suitable bounded-increment assumptions,

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_0].$$

The assumptions cannot be dropped. A fair game can be turned into an apparently guaranteed win only by allowing strategies with unbounded losses, unbounded stopping times, or non-integrable payoffs.

2.4.1.2 Doob convergence theorem

A suitably bounded submartingale or supermartingale converges almost surely and often in L^1. This theorem formalizes the idea that a sequence of improving conditional forecasts can settle to a limit.

2.4.1.3 Martingale representation theorem

Under the filtration generated by Brownian motion, every square-integrable martingale can be represented as

$$M_t = M_0 + \int_0^t H_s \, \mathrm{d}W_s$$

for an adapted square-integrable process H. In a complete Brownian market, the integrand corresponds to a hedging strategy. This theorem is one reason Brownian market models can replicate a broad class of claims [3, 5].

2.4.1.4 Fundamental theorem of asset pricing

In a finite-dimensional or suitably regular continuous-time market, absence of arbitrage is linked to the existence of an equivalent measure under which discounted tradable prices are local martingales. Market completeness is linked to uniqueness of that equivalent martingale measure. Precise statements depend on the market class and admissibility definitions [8, 9].

2.5 Markov processes and transition kernels

A process is Markov if, conditionally on the present state, the future is independent of the past:

$$\mathbb{P}(X_t \in A \mid \mathcal{F}_s) = \mathbb{P}(X_t \in A \mid X_s), \qquad s < t.$$

The transition kernel $K_{s,t}(x, A)$ gives the probability of reaching A at time t from state x at time s.

Markovianity is computationally powerful because a value function can often be written as $V(t, X_t)$ rather than as a functional of the whole past. A path-dependent claim can sometimes be made Markov by adding state variables. For an Asian option, one may augment the state with the running integral $A_t = \int_0^t S_u \, \mathrm{d}u$.

2.6 Shocks: continuous noise and jumps

The word *shock* can refer to two mathematically different mechanisms.

- A Brownian shock is infinitesimal and continuous at the path level. Over Δt, it has size $O(\sqrt{\Delta t})$.
- A jump shock changes the state discontinuously, often modeled by a Poisson process or a random jump measure.

A jump-diffusion may be written schematically as

$$\frac{\mathrm{d}S_t}{S_{t^-}} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t + (J-1)\,\mathrm{d}N_t,$$

where N_t counts jumps and J is a random jump multiplier. Jumps can represent earnings announcements, defaults, policy surprises, or market gaps. They introduce additional incompleteness because jump risk may not be hedgeable by the underlying alone.

2.7 Particle systems

A particle system represents a probability distribution by a finite cloud of states $X_t^1, \ldots, X_t^N$. Its empirical measure is

$$\mu_t^N = \frac{1}{N}\sum_{i=1}^{N} \delta_{X_t^i}.$$

For independent particles, μ_t^N approximates the common law by the law of large numbers. In an interacting system, each particle may depend on the empirical distribution:

$$\mathrm{d}X_t^i = b(X_t^i, \mu_t^N)\,\mathrm{d}t + \sigma(X_t^i, \mu_t^N)\,\mathrm{d}W_t^i.$$

Such systems arise in mean-field models, particle filters, sequential Monte Carlo, optimal transport, and numerical approximations of nonlinear Fokker-Planck equations [18, 19].

Experiment: Mean-field attraction

The simulated particles start in two separated clusters. Each particle is pulled both toward zero and toward the empirical mean while receiving independent Brownian noise. The cloud contracts toward a common distribution. The final empirical mean was 0.02385 and the final empirical standard deviation was 0.42859.

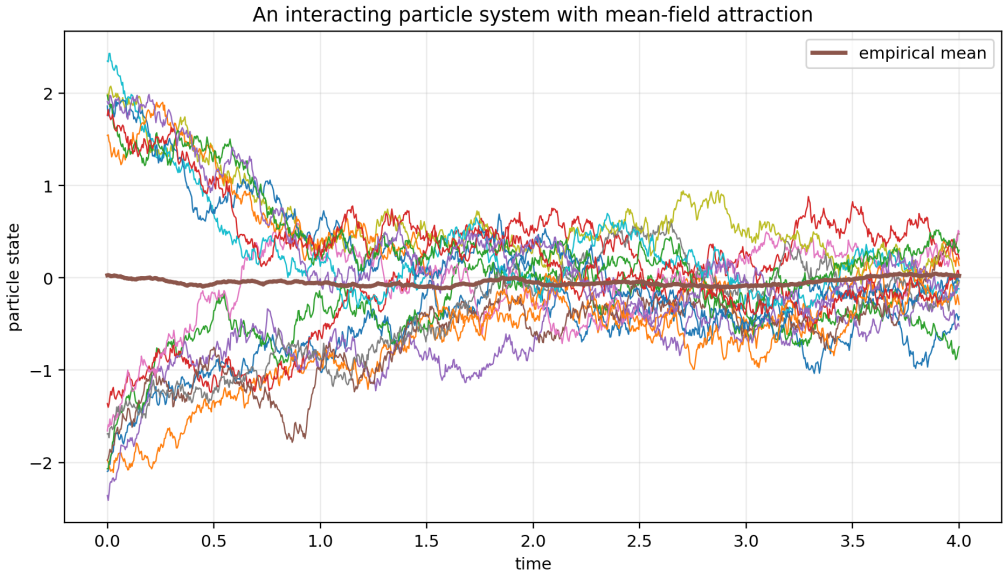


Figure 2.3: Selected trajectories from an interacting particle system. The thick curve is the empirical mean.

Chapter summary

- A stochastic process is a time-indexed family of random variables; a path is one realization through time.
- A filtration represents information, and adaptedness rules out future-looking strategies.
- Brownian motion is continuous, nowhere classically differentiable, and has quadratic variation $[W]_T = T$.
- Martingales model zero conditional drift; optional stopping, convergence, representation, and asset-pricing theorems serve different roles.
- Shocks may be continuous diffusive increments or discontinuous jumps.
- Particle systems approximate distributions and can include interactions through the empirical measure.

Chapter 3

Stochastic calculus, Itô's isometry, Itô's lemma, and SDEs

3.1 Why ordinary calculus is not enough

For a differentiable path x_t, the increment of $f(t, x_t)$ is approximated by

$$\mathrm{d}f = f_t\,\mathrm{d}t + f_x\,\mathrm{d}x.$$

Brownian increments satisfy $\mathrm{d}W = O(\sqrt{\mathrm{d}t})$, so $(\mathrm{d}W)^2 = O(\mathrm{d}t)$ and cannot be discarded. This produces an additional second-derivative term.

The core of Itô calculus is therefore not mysterious notation. It is the systematic accounting of nonzero quadratic variation.

3.2 The Itô integral

Begin with a simple adapted process

$$H_t = \sum_{i=0}^{n-1} H_i \mathbf{1}_{(t_i, t_{i+1}]}(t), \qquad H_i \in \mathcal{F}_{t_i}.$$

Its Itô integral is

$$\int_0^T H_t\,\mathrm{d}W_t = \sum_{i=0}^{n-1} H_i (W_{t_{i+1}} - W_{t_i}).$$

The left-endpoint measurability encodes non-anticipation. General square-integrable integrands are obtained by completing these simple processes in an L^2 norm.

3.2.1 Itô's isometry

Itô isometry

For an adapted process H satisfying $\mathbb{E}\int_0^T H_t^2\,\mathrm{d}t < \infty$,

$$\mathbb{E}\left[\left(\int_0^T H_t\,\mathrm{d}W_t\right)^2\right] = \mathbb{E}\left[\int_0^T H_t^2\,\mathrm{d}t\right].$$

This identity turns the stochastic integral into a continuous linear map from a space of integrands to $L^2(\Omega)$. It is the key construction tool and also implies that the integral has mean zero when integrable.

For $H_t = t$ on $[0,1]$,

$$\mathbb{E}\left[\left(\int_0^1 t\,\mathrm{d}W_t\right)^2\right] = \int_0^1 t^2\,\mathrm{d}t = \frac{1}{3}.$$

Experiment: Numerical Itô isometry

The integral was approximated by left-endpoint sums over 60,000 simulated paths. With 512 time steps, the estimated second moment was 0.33268, compared with the exact value $1/3 \approx 0.33333$.

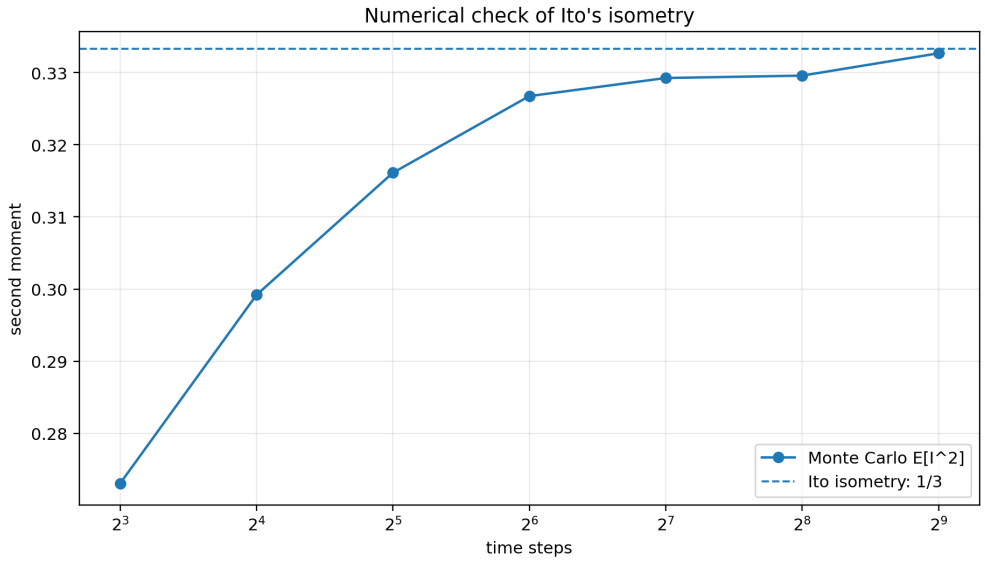


Figure 3.1: The simulated second moment converges toward the deterministic integral predicted by Itô's isometry.

3.3 Itô's lemma

Suppose

$$\mathrm{d}X_t = a(t, X_t)\,\mathrm{d}t + b(t, X_t)\,\mathrm{d}W_t$$

and f is sufficiently smooth. Then

$$\boxed{\mathrm{d}f(t, X_t) = \left(f_t + a f_x + \frac{1}{2} b^2 f_{xx}\right)\,\mathrm{d}t + b f_x\,\mathrm{d}W_t.}$$

The term $\frac{1}{2} b^2 f_{xx}\,\mathrm{d}t$ is the Itô correction.

A useful mnemonic is the multiplication table

$$(\mathrm{d}t)^2 = 0, \qquad \mathrm{d}t\,\mathrm{d}W_t = 0, \qquad (\mathrm{d}W_t)^2 = \mathrm{d}t,$$

understood as shorthand for orders of variation, not literal infinitesimals.

3.3.1 Multidimensional form

If $X_t \in \mathbb{R}^d$ satisfies

$$\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + \Sigma(t, X_t)\,\mathrm{d}W_t,$$

where $W_t \in \mathbb{R}^m$, then

$$\mathrm{d}f = \left(f_t + b^\top \nabla f + \frac{1}{2} \operatorname{tr}\left(\Sigma\Sigma^\top \nabla^2 f \right) \right) \mathrm{d}t + (\nabla f)^\top \Sigma \, \mathrm{d}W_t.$$

The operator

$$\mathcal{L}f = b^\top \nabla f + \frac{1}{2} \operatorname{tr}(\Sigma\Sigma^\top \nabla^2 f)$$

is the infinitesimal generator of the diffusion.

3.4 Stochastic differential equations

An SDE

$$\mathrm{d}X_t = b(t, X_t) \, \mathrm{d}t + \sigma(t, X_t) \, \mathrm{d}W_t, \qquad X_0 = x_0,$$

is interpreted through its integral form

$$X_t = x_0 + \int_0^t b(s, X_s) \, \mathrm{d}s + \int_0^t \sigma(s, X_s) \, \mathrm{d}W_s.$$

A *strong solution* is constructed on a given probability space from a given Brownian motion. A *weak solution* allows the probability space and Brownian motion to be part of the solution. Strong uniqueness asks whether two solutions driven by the same Brownian path coincide; uniqueness in law asks only whether they have the same distribution.

A standard sufficient condition for a unique strong solution is that b and σ are globally Lipschitz in the state and satisfy linear growth bounds:

$$|b(t,x) - b(t,y)| + |\sigma(t,x) - \sigma(t,y)| \le L \, |x - y| \, ,$$

$$|b(t,x)| + |\sigma(t,x)| \le C(1 + |x|).$$

These are sufficient, not necessary, conditions [4, 3].

3.5 Geometric Brownian motion

Geometric Brownian motion

A geometric Brownian motion (GBM) satisfies

$$\mathrm{d}S_t = \mu S_t \, \mathrm{d}t + \sigma S_t \, \mathrm{d}W_t, \qquad S_0 > 0.$$

Apply Itô's lemma to $f(S) = \log S$. Since $f'(S) = 1/S$ and $f''(S) = -1/S^2$,

$$\begin{aligned} \mathrm{d} \log S_t &= \frac{1}{S_t} \, \mathrm{d}S_t - \frac{1}{2} \frac{1}{S_t^2} (\, \mathrm{d}S_t)^2 \\ &= \left(\mu - \frac{1}{2}\sigma^2 \right) \mathrm{d}t + \sigma \, \mathrm{d}W_t. \end{aligned}$$

Integrating gives the explicit solution

$$\boxed{S_t = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2 \right) t + \sigma W_t \right].}$$

Hence $S_t > 0$ almost surely and $\log S_t$ is Gaussian. The terminal price is lognormal, with

$$\mathbb{E}[S_t] = S_0 e^{\mu t}, \qquad \text{median}(S_t) = S_0 e^{(\mu - \sigma^2/2)t}.$$

The difference between the mean and median is a direct effect of lognormal skewness.

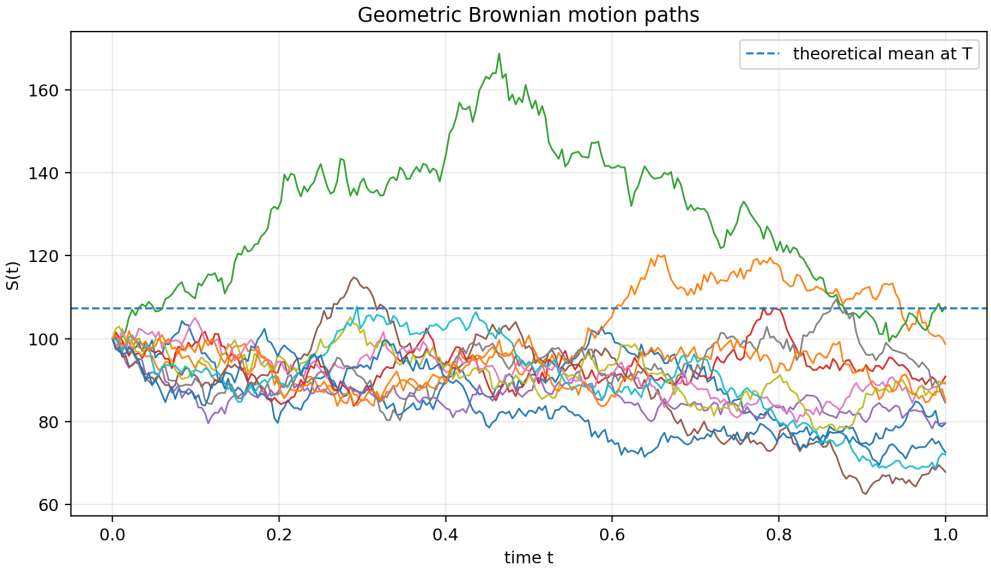


Figure 3.2: GBM paths remain positive but exhibit multiplicative dispersion.

Experiment: GBM distribution

With $S_0 = 100$, $\mu = 7\%$, $\sigma = 25\%$, and $T = 1$, 120,000 exact terminal samples had mean 107.3796. The theoretical mean is $100e^{0.07} = 107.2508$. The histogram agrees with the lognormal density.

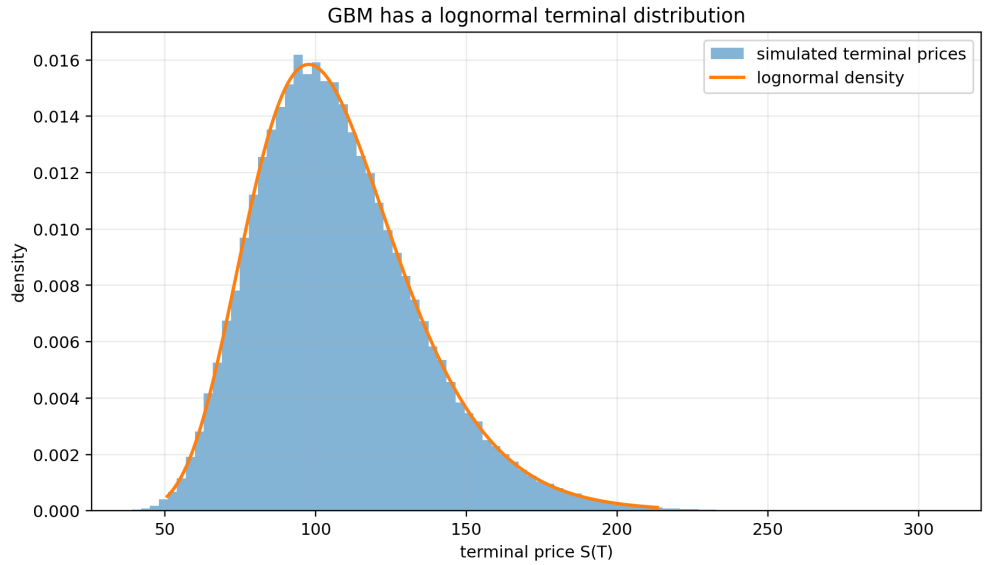


Figure 3.3: Exact GBM simulation and its lognormal terminal density.

3.6 Euler-Maruyama discretization

For a grid $t_n = n\Delta t$, the Euler-Maruyama scheme is

$$X_{n+1} = X_n + b(t_n, X_n)\Delta t + \sigma(t_n, X_n)\sqrt{\Delta t}\, Z_n, \qquad Z_n \sim \mathcal{N}(0,1).$$

Under standard conditions, the strong error is typically order $1/2$:

$$\mathbb{E}\left[\left|X_T - X_T^{\Delta t}\right|^2\right]^{1/2} = O(\Delta t^{1/2}).$$

The weak error for sufficiently smooth test functions is often order 1:

$$\left|\mathbb{E}[f(X_T)] - \mathbb{E}[f(X_T^{\Delta t})]\right| = O(\Delta t).$$

Strong accuracy matters for pathwise coupling and barrier monitoring. Weak accuracy matters when only expected payoffs are required.

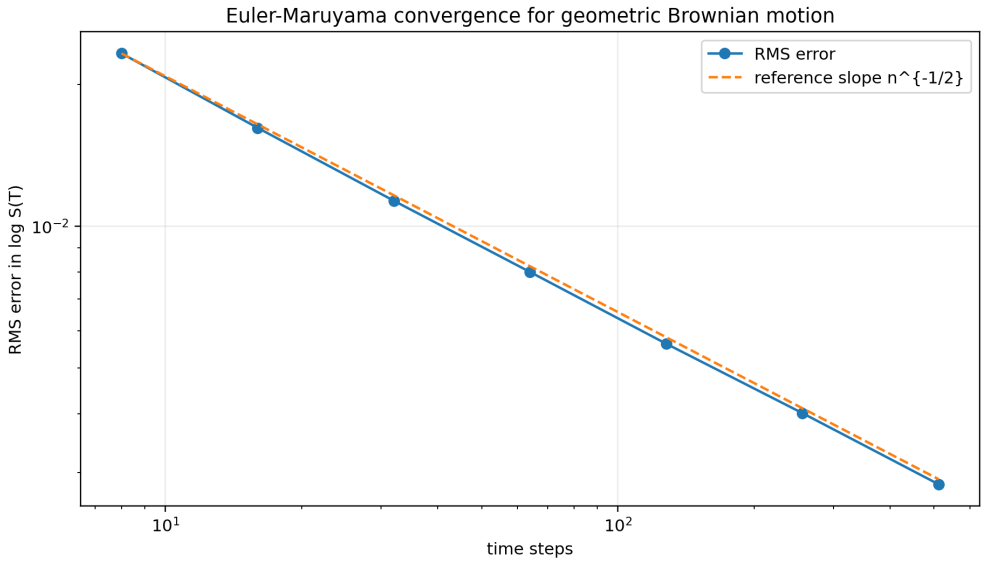


Figure 3.4: The Euler-Maruyama strong error for GBM is consistent with the reference $n^{-1/2}$ rate.

3.7 Itô versus Stratonovich

The Stratonovich integral, denoted $\circ\, \mathrm{d}W_t$, uses a symmetric limiting convention and obeys the ordinary chain rule. The conversion

$$\mathrm{d}X_t = b_{\mathrm{Strat}}(X_t)\,\mathrm{d}t + \sigma(X_t) \circ \mathrm{d}W_t$$

corresponds in one dimension to the Itô SDE

$$\mathrm{d}X_t = \left[b_{\mathrm{Strat}}(X_t) + \frac{1}{2}\sigma(X_t)\sigma'(X_t)\right]\mathrm{d}t + \sigma(X_t)\,\mathrm{d}W_t.$$

Finance usually uses Itô calculus because adapted left-endpoint integrands align with non-anticipating trading. Stratonovich calculus is often natural in physical systems obtained as limits of smooth noise.

3.8 Fokker-Planck and backward equations

If X_t has density $p(t,x)$ and generator $\mathcal{L}$, the density evolves under the forward Kolmogorov or Fokker-Planck equation

$$\partial_t p = -\nabla \cdot (bp) + \frac{1}{2}\sum_{i,j} \partial_{ij}(a_{ij}p), \qquad a = \Sigma\Sigma^\top.$$

Expected future payoffs solve the backward equation. If

$$u(t,x) = \mathbb{E}[g(X_T) \mid X_t = x],$$

then, under regularity,

$$\partial_t u + \mathcal{L}u = 0, \qquad u(T,x) = g(x).$$

The Black-Scholes PDE is a discounted version of this backward equation.

Chapter summary

- Itô integration is built from non-anticipating sums and completed using Itô's isometry.
- Itô's lemma adds a second-derivative correction because Brownian motion has nonzero quadratic variation.
- SDEs are integral equations; strong versus weak solutions and pathwise versus distributional uniqueness are distinct.
- GBM has an explicit positive solution and a lognormal terminal law.
- Euler-Maruyama typically has strong order 1/2 and weak order 1 under regularity.
- The generator links SDEs to backward pricing equations and forward density equations.

Chapter 4

Option contracts: strikes, moneyness, exercise styles, and path dependence

4.1 The anatomy of an option

An option is a contract whose cash flow depends on an underlying state such as an equity price, rate, exchange rate, commodity price, or index. The key ingredients are:

- the underlying process S_t;
- the strike price K;
- the maturity T;
- the payoff rule;
- the permitted exercise times;
- any path-monitoring, barriers, averages, resets, or rebates.

Strike price

The strike price K is the contractual price at which the holder may buy or sell the underlying, or the reference level used to compute a cash-settled payoff.

At maturity, a standard European call and put have payoffs

$$(S_T - K)^+ = \max(S_T - K, 0), \qquad (K - S_T)^+ = \max(K - S_T, 0).$$

The plus notation $x^+ = \max(x, 0)$ is widely used in derivatives.

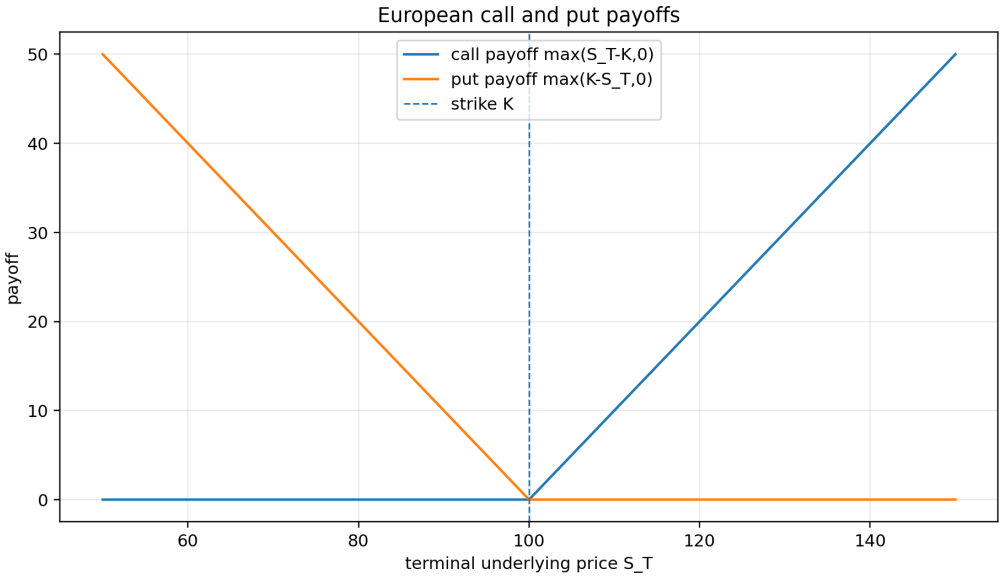


Figure 4.1: Call and put payoffs as functions of the terminal underlying price.

4.2 Call versus put

A *call* provides upside exposure: its payoff increases when the underlying finishes above the strike. A *put* provides downside exposure: its payoff increases when the underlying finishes below the strike.

The buyer pays a premium and has a right but not an obligation. The seller receives the premium and has the corresponding contingent obligation. The option's value before maturity is not merely its current intrinsic payoff. It also contains time value arising from the possibility of favorable future movement.

4.2.1 Put-call parity

For European options with common strike and maturity and deterministic rates/dividends,

$$C_t - P_t = S_t e^{-q\tau} - K e^{-r\tau}, \qquad \tau = T - t.$$

The equality follows because both sides have the same terminal payoff after financing and dividends are accounted for. It is a no-arbitrage relation independent of Black-Scholes dynamics.

4.3 Moneyness

Moneyness measures where the strike lies relative to a relevant underlying or forward level. Several conventions coexist.

Convention	Formula	Interpretation
Spot moneyness	S_t/K or K/S_t	compares strike with current spot; always state the direction

Forward moneyness	$F_{t,T}/K$	removes deterministic carry to compare strike with the forward
Log-moneyness	$k = \log(K/F_{t,T})$	dimensionless and symmetric under multiplicative scaling
Standardized moneyness	$\log(F/K)/(\sigma\sqrt{\tau})$	measures distance from the strike in diffusion standard deviations

For a call:

- in the money: $S_t > K$ under spot convention;
- at the money: $S_t \approx K$, or often $F_{t,T} \approx K$ for quoted implied volatility;
- out of the money: $S_t < K$.

For a put, the in-the-money and out-of-the-money inequalities reverse.

Moneyness conventions are not universal

Some desks quote K/F and others quote F/K; some use $\log(K/F)$ and others $\log(F/K)$. A number called "moneyness" is incomplete unless the convention is stated.

4.4 European, American, and Bermudan exercise

4.4.1 European options

A European option may be exercised only at maturity. Its discounted payoff is a single terminal random variable, which makes risk-neutral expectation particularly direct.

4.4.2 American options

An American option may be exercised at any admissible time up to maturity. Its value is an optimal stopping problem:

$$V_t = \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^\tau r_u \, \mathrm{d}u} h(\tau, S_\tau) \,\middle|\, \mathcal{F}_t \right],$$

where $\mathcal{T}_{t,T}$ is the set of stopping times between t and T.

For a non-dividend-paying stock with nonnegative rates, early exercise of a standard American call is not optimal: exercising sacrifices remaining optionality and pays the strike early. Early exercise of a put can be optimal because receiving the strike sooner has financing value. Dividends, borrow costs, negative rates, and contract details can alter these conclusions.

4.4.3 Bermudan options

A Bermudan option may be exercised only on a specified finite set of dates. It lies between European and American exercise and is common in callable rate products.

4.5 What makes an option exotic?

An exotic option has a payoff or exercise rule beyond a standard vanilla call or put. "Exotic" does not necessarily mean rare; many exotics are standard in particular markets.

Contract	Core path functional
Asian	arithmetic or geometric average, such as $T^{-1}\int_0^T S_t\,\mathrm{d}t$
Lookback	running maximum or minimum
Digital	indicator $\mathbf{1}_{\{S_T>K\}}$ or a cash/asset-or-nothing variant
Barrier	whether the path crosses a level before maturity
Basket	weighted combination of several underlyings
Cliquet	sum of capped or floored returns over reset periods
Reset option	strike or another contract parameter changes according to a rule
Autocallable	coupons and early redemption triggered on observation dates

4.6 Barrier options

A barrier option activates or terminates when the underlying crosses a barrier. Common labels are:

- knock-in versus knock-out;
- up barrier versus down barrier;
- continuously monitored versus discretely monitored;
- with or without a rebate.

An up-and-out call with barrier $B > S_0$ and zero rebate has payoff

$$(S_T - K)^+ \mathbf{1}_{\{\max_{0\le t\le T} S_t < B\}}.$$

Two paths with the same terminal value can produce different payoffs because one may have touched the barrier and the other may not.

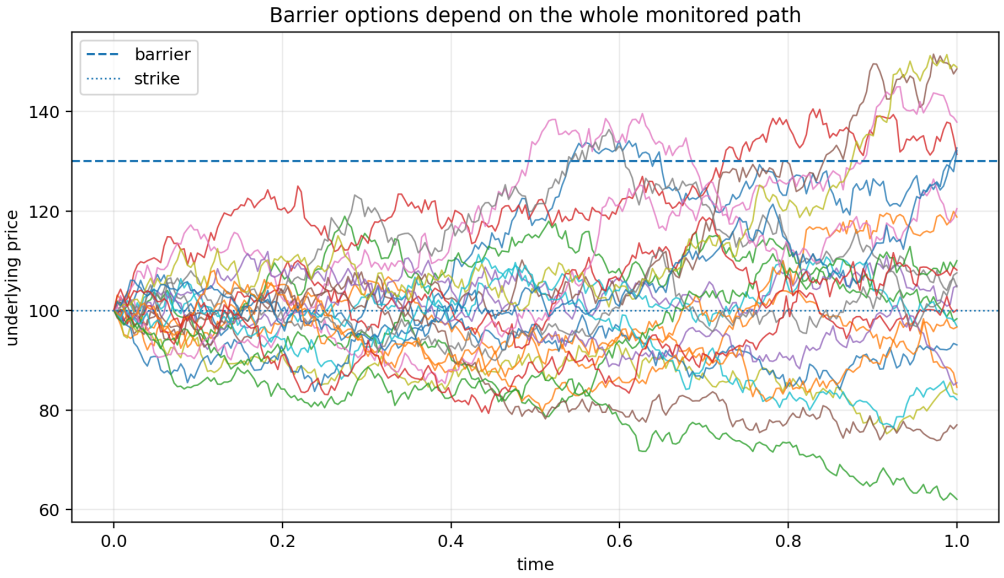


Figure 4.2: The terminal value alone does not determine a barrier payoff.

Experiment: A discretely monitored up-and-out call

Under risk-neutral GBM with $S_0 = K = 100$, barrier $B = 130$, $r = 3\%$, $\sigma = 22\%$, one year, and 252 monitoring dates, 300,000 simulated paths gave a price of approximately 2.9611. About 77.43% of paths survived without crossing the barrier. A finer monitoring grid would generally reduce the value because more crossings are detected.

For compatible zero-rebate contracts, a knock-in/knock-out parity often holds:

$$\text{knock-in} + \text{knock-out} = \text{vanilla}.$$

The precise parity depends on matching strike, barrier, maturity, monitoring, and rebate conventions.

4.7 Reset contracts

A reset contract changes a parameter when a stated event occurs or at scheduled dates. Examples include:

- a resettable strike K_t linked to the running minimum or an observation-date spot;
- a ratchet or cliquet that locks in periodic gains;
- a convertible bond whose conversion terms reset;
- a swap whose coupon or spread resets from a reference rate.

A simple reset put might replace the initial strike by

$$K_{\text{reset}} = \max(K_0, S_{t_r})$$

at reset date t_r. The final payoff is $(K_{\text{reset}} - S_T)^+$. Because the strike becomes random and path-dependent, the contract is not equivalent to a fixed-strike vanilla put.

4.8 Path dependence and state augmentation

A payoff is path-dependent when it depends on more than the terminal state. Formally, it is a functional

$$H = \Phi\big((S_t)_{0\le t\le T}\big).$$

Examples include barriers, averages, occupation times, drawdowns, realized variance, and running extrema.

Some path dependence can be compressed into additional state variables:

$$A_t = \int_0^t S_u \,\mathrm{d}u \qquad \text{for an Asian payoff,}$$

$$M_t = \max_{0\le u\le t} S_u \qquad \text{for a lookback or barrier,}$$

$$V_t = \int_0^t \sigma_u^2 \,\mathrm{d}u \qquad \text{for realized-variance claims.}$$

Then (S_t, A_t) or (S_t, M_t) may be Markov even when S_t alone is insufficient.

Not every path functional has a convenient finite-dimensional state representation. Signature methods, recurrent networks, and functional regression provide alternatives for representing richer path information.

4.9 A contract is not yet a model

The payoff rule specifies what is paid. A pricing model specifies a law for market states and a no-arbitrage valuation rule. The same barrier option can receive different model prices under GBM, local volatility, stochastic volatility, or jump dynamics even when all models fit a subset of vanilla prices.

Terminal calibration does not guarantee path calibration

Vanilla European prices constrain terminal marginal distributions under the pricing measure. Barrier, Asian, and early-exercise products depend on joint path behavior. Two models can match every quoted vanilla price in a grid and still disagree materially on an exotic payoff.

Chapter summary

- The strike is the contract's reference transaction or payoff level.
- Moneyness is a dimensionless comparison of strike with spot or forward; conventions must be stated.
- Calls benefit from upside and puts from downside; put-call parity is a no-arbitrage identity.
- European, American, and Bermudan options differ by exercise opportunities.
- Barrier and reset contracts are path-dependent and require precise monitoring rules.
- State augmentation can turn some path-dependent problems into higher-dimensional Markov problems.

Chapter 5

No-arbitrage, risk-neutral simulation, and Black-Scholes

5.1 Prices from replication rather than prediction

A derivative price is not generally obtained by forecasting the most likely asset path under the physical measure. In an arbitrage-free complete model, the price is determined by the cost of a self-financing trading strategy that replicates the payoff.

The physical drift μ describes expected return under $\mathbb{P}$. The risk-neutral drift is tied to financing and carry. The separation is central:

$$\text{statistical prediction under } \mathbb{P} \quad \neq \quad \text{arbitrage-free valuation under } \mathbb{Q}.$$

5.2 Numeraire and risk-neutral measure

Let B_t be the money-market account,

$$B_t = \exp\left(\int_0^t r_u \, \mathrm{d}u\right).$$

A probability measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ is a risk-neutral or equivalent martingale measure for this numeraire if discounted tradable gains are local martingales under $\mathbb{Q}$.

For a replicable payoff H paid at T,

$$\boxed{V_t = B_t \mathbb{E}^{\mathbb{Q}}\left[\frac{H}{B_T}\,\middle|\,\mathcal{F}_t\right].}$$

With constant rate r,

$$V_t = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[H \mid \mathcal{F}_t].$$

5.2.1 Why the drift changes

Suppose under $\mathbb{P}$,

$$\frac{\mathrm{d}S_t}{S_t} = (\mu - q)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{P}}.$$

Under suitable conditions, Girsanov's theorem defines a new Brownian motion

$$\mathrm{d}W_t^{\mathbb{Q}} = \mathrm{d}W_t^{\mathbb{P}} + \lambda_t\,\mathrm{d}t, \qquad \lambda_t = \frac{\mu - r}{\sigma},$$

so that

$$\frac{\mathrm{d}S_t}{S_t} = (r - q)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}.$$

The stochastic paths are reweighted, not physically altered. Equivalent measures agree on which events have zero probability but assign different probabilities to non-null events.

5.3 Fundamental theorem and completeness

The fundamental theorem of asset pricing links no-arbitrage to equivalent martingale measures. In broad semimartingale settings, the exact no-arbitrage condition is more technical than the elementary "no free lunch" phrase, but the conceptual message is stable:

- existence of a suitable equivalent martingale measure corresponds to absence of economically meaningful arbitrage;
- uniqueness of that measure corresponds, under appropriate assumptions, to market completeness.

If several martingale measures are compatible with traded prices, an unhedgeable claim need not have a unique no-arbitrage price. Additional modeling, preference, or risk-premium assumptions are then required [8, 9].

5.4 Black-Scholes dynamics

The Black-Scholes model assumes under $\mathbb{Q}$

$$\mathrm{d}S_t = (r-q)S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t^{\mathbb{Q}},$$

with constant r, q, σ. Its exact solution from time t to T is

$$S_T = S_t \exp\left[\left(r - q - \frac{1}{2}\sigma^2\right)\tau + \sigma\sqrt{\tau}\,Z\right], \qquad Z \sim \mathcal{N}(0,1),$$

where $\tau = T - t$.

5.5 Deriving the Black-Scholes PDE

Let $V(t,S)$ be a smooth derivative value. Itô's lemma gives

$$\mathrm{d}V = \left(V_t + (r-q)SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS}\right)\mathrm{d}t + \sigma S V_S\,\mathrm{d}W^{\mathbb{Q}}.$$

Construct a locally delta-hedged portfolio with one derivative and $-V_S$ units of stock. The Brownian term cancels. A locally riskless self-financing portfolio must earn the risk-free rate. Accounting for dividends yields

$$\boxed{V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + (r-q)SV_S - rV = 0.}$$

For a European payoff $h(S_T)$, the terminal condition is $V(T,S) = h(S)$.

The same equation follows from the Feynman-Kac formula applied to the risk-neutral diffusion. Delta hedging and conditional expectation are therefore two views of the same pricing structure.

5.6 Black-Scholes call and put formulas

For a European call,

$$C = Se^{-q\tau}N(d_1) - Ke^{-r\tau}N(d_2),$$
$$d_1 = \frac{\log(S/K) + (r - q + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}},$$
$$d_2 = d_1 - \sigma\sqrt{\tau}.$$

For a put,

$$P = Ke^{-r\tau}N(-d_2) - Se^{-q\tau}N(-d_1).$$

Here N is the standard normal cumulative distribution function.

Under the risk-neutral measure, $N(d_2) = \mathbb{Q}(S_T > K \mid S_t = S)$ in this model. The factor $e^{-q\tau}N(d_1)$ is the call Delta and can be interpreted through a stock-numeraire measure; it is not simply the same probability with a different threshold.

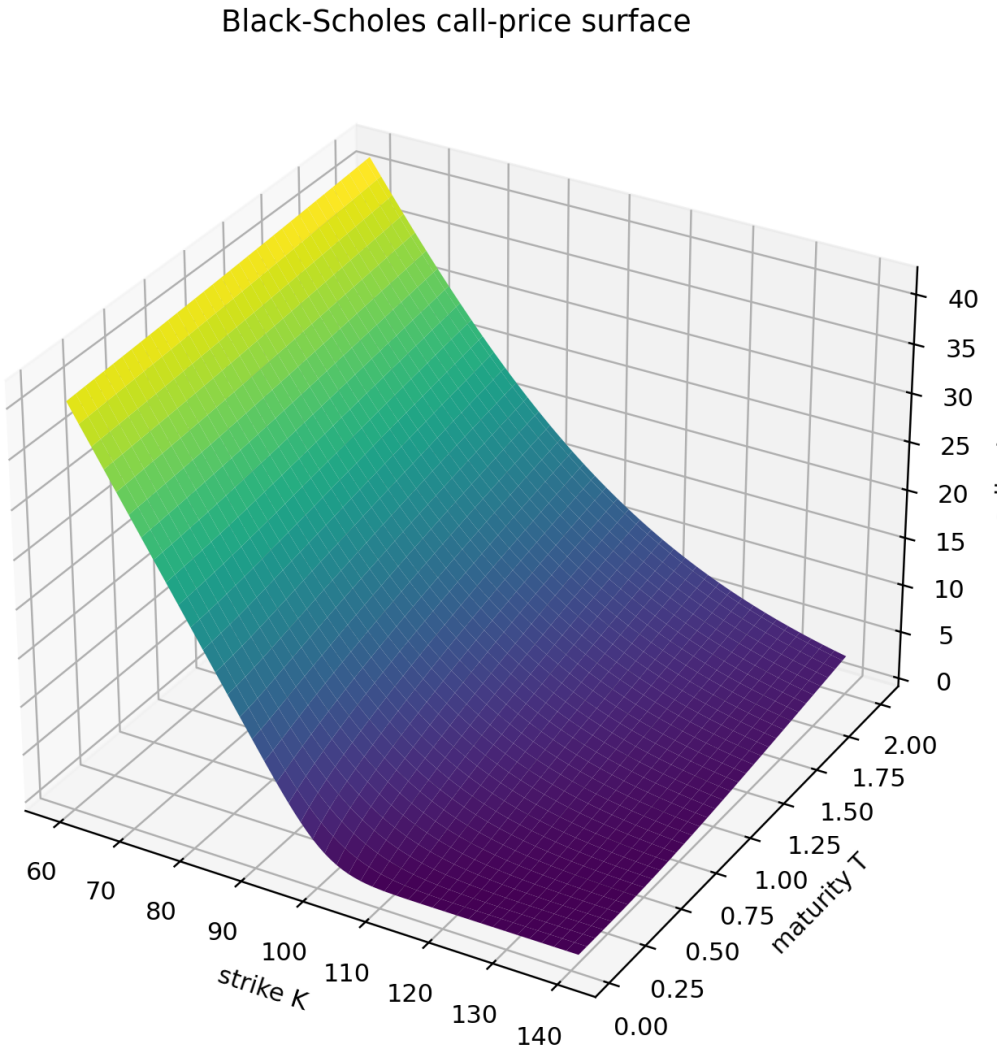


Figure 5.1: Black-Scholes call prices across strike and maturity for fixed spot, rates, dividends, and volatility.

5.7 Risk-neutral simulation

A Monte Carlo price uses simulated paths under $\mathbb{Q}$:

$$\widehat{V}_N = e^{-rT}\frac{1}{N}\sum_{i=1}^{N} H^{(i)}.$$

If the discounted payoff has finite variance, then

$$\sqrt{N}(\widehat{V}_N - V) \Longrightarrow \mathcal{N}(0, \sigma_H^2),$$

so standard error decreases only as $N^{-1/2}$.

For a European GBM payoff, exact terminal simulation avoids time-discretization error. For barriers, American exercise, stochastic volatility, or path functionals, a time grid is usually required.

Listing 5.1: Risk-neutral Monte Carlo for a European call

```
import numpy as np

z = rng.normal(size=n_paths)
ST = S0 * np.exp((r - q - 0.5*sigma**2)*T
                 + sigma*np.sqrt(T)*z)
payoff = np.maximum(ST - K, 0.0)
price = np.exp(-r*T) * payoff.mean()
standard_error = np.exp(-r*T) * payoff.std(ddof=1) / np.sqrt(n_paths)
```

Experiment: Monte Carlo price and convergence

For $S_0 = K = 100$, $r = 3\%$, $\sigma = 22\%$, $T = 1$, 400,000 exact risk-neutral paths produced 10.2162 with estimated standard error 0.0247. The exact Black-Scholes price is 10.1871, a difference of about 1.18 estimated standard errors. Repeated experiments display the expected $N^{-1/2}$ convergence rate.

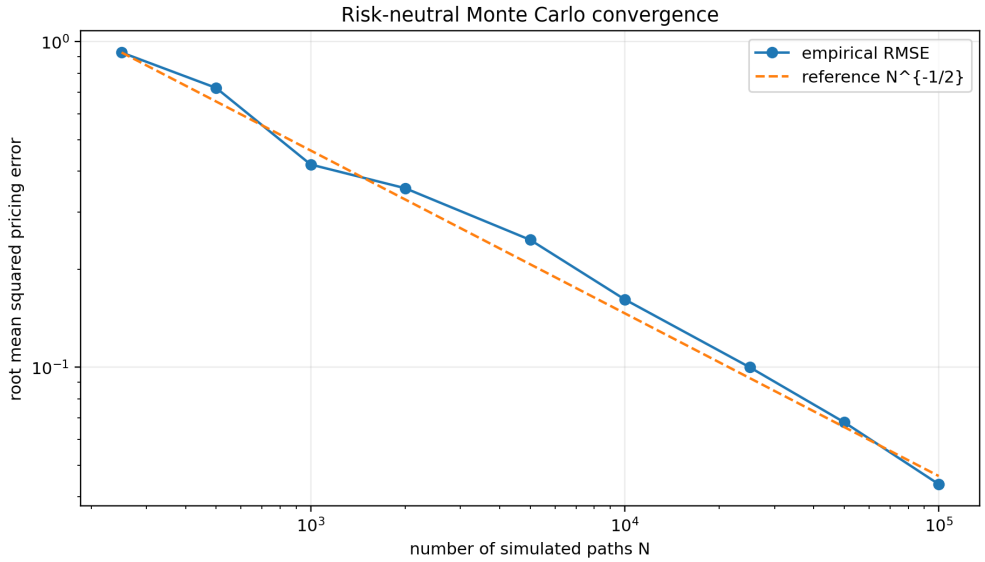


Figure 5.2: Monte Carlo root mean squared error follows the characteristic inverse-square-root rate.

5.8 Variance reduction

Because brute-force convergence is slow, simulation often uses:

- antithetic variables, pairing Z with $-Z$;
- control variates with known expectations;
- importance sampling for rare events;
- stratification, Latin hypercube, or quasi-Monte Carlo;
- conditional Monte Carlo, analytically integrating part of the randomness;
- multilevel Monte Carlo, combining grids of different resolutions.

A method must reduce variance without introducing unquantified bias.

5.9 Black-Scholes assumptions and what fails

The model assumes continuous trading, frictionless markets, lognormal diffusion, constant volatility, and a sufficiently liquid underlying. Real markets show transaction costs, jumps, stochastic rates, changing volatility, liquidity effects, funding asymmetries, and discrete hedging.

The formula remains foundational because it provides:

- a benchmark and quoting convention;
- an invertible map from price to implied volatility;
- analytic Greeks;
- a local approximation inside richer models;
- a clean example of replication, PDEs, and risk-neutral expectation.

A correct price inside a model is conditional on the model

No-arbitrage removes some degrees of freedom, but it does not certify that the assumed dynamics describe every relevant risk. Model risk is the gap between internal mathematical consistency and external market adequacy.

Chapter summary

- Risk-neutral valuation prices discounted payoffs under a measure where discounted tradables are martingales.
- The physical drift is not generally needed to price a replicable derivative.
- Delta hedging and Feynman-Kac lead to the same Black-Scholes PDE.
- The closed-form call and put formulas follow from a lognormal risk-neutral terminal law.
- Monte Carlo is flexible but converges at $N^{-1/2}$ without additional structure.
- Market completeness is tied to uniqueness of the martingale measure and replicability of claims.

Chapter 6

Volatility, implied volatility, local volatility, and Dupire inversion

6.1 What volatility measures

In a diffusion

$$\frac{\mathrm{d}S_t}{S_t} = \mu_t \,\mathrm{d}t + \sigma_t \,\mathrm{d}W_t,$$

σ_t is the instantaneous standard deviation rate of returns. Over a short interval Δt, the diffusive return standard deviation is approximately

$$\sigma_t \sqrt{\Delta t}.$$

Volatility therefore has units of inverse square-root time. A quoted annualized volatility of 20% means a one-year diffusion standard deviation of roughly 20%, not a deterministic annual move.

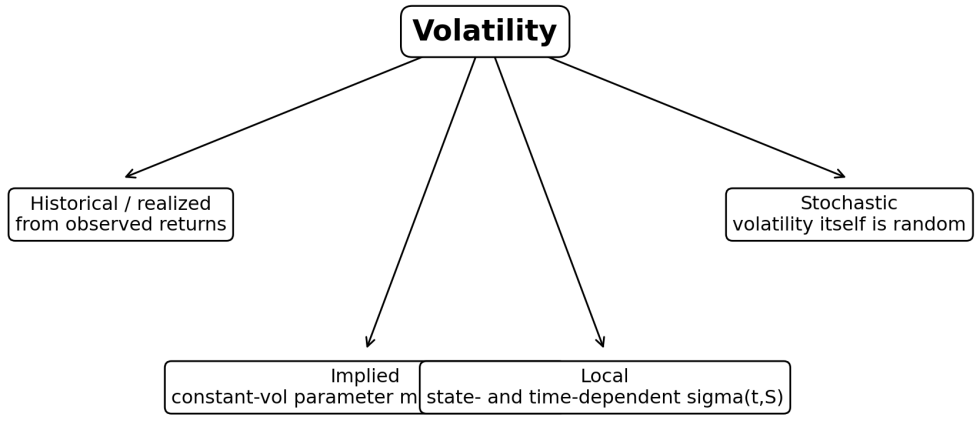


Figure 6.1: The word volatility refers to several distinct mathematical objects.

6.2 Realized and historical volatility

For log returns $r_i = \log(S_{t_i}/S_{t_{i-1}})$ sampled at spacing Δt, a basic realized-variance estimate is

$$\widehat{\sigma}^2_{\text{ann}} = \frac{1}{n\Delta t}\sum_{i=1}^{n}(r_i - \bar{r})^2$$

under a simple stationary approximation. High-frequency realized variance often uses $\sum r_i^2$ over a day and must confront microstructure noise, jumps, asynchronous observations, and market closures.

Historical volatility is backward-looking and measure-dependent: it estimates variability under the data-generating physical measure, not directly under the risk-neutral measure.

6.3 Implied volatility

Black-Scholes implied volatility

Given a market option price C^{mkt}, the implied volatility σ_{imp} is the value satisfying

$$C^{\text{BS}}(S, K, T, r, q, \sigma_{\text{imp}}) = C^{\text{mkt}}.$$

For a non-degenerate European option, Black-Scholes Vega is positive:

$$\nu = \frac{\partial C}{\partial \sigma} = Se^{-q\tau}\phi(d_1)\sqrt{\tau} > 0.$$

Therefore the Black-Scholes price is strictly increasing in volatility between its no-arbitrage bounds, so the implied volatility is unique when it exists.

Implied volatility is not the claim that future volatility will be constant. It is a re-parameterization of one option price through a benchmark model.

6.4 The implied-volatility surface

Market options at different strikes and maturities generally produce different implied volatilities:

$$(K, T) \longmapsto \sigma_{\text{imp}}(K, T).$$

A cross-section at fixed maturity may show a smile or skew. Equity index options often exhibit higher implied volatility for lower strikes, reflecting downside-tail pricing and leverage-like effects. The exact shape varies across assets and time.

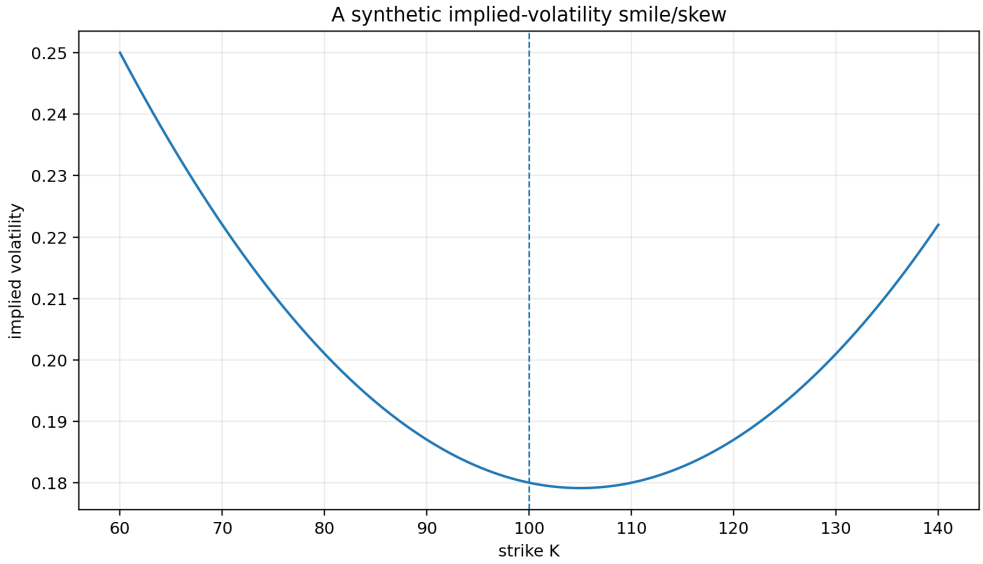


Figure 6.2: A synthetic strike-dependent implied-volatility smile with skew.

A valid surface must correspond to option prices without static arbitrage. For each maturity, call prices should be decreasing and convex in strike:

$$\partial_K C \le 0, \qquad \partial_{KK} C \ge 0.$$

The second derivative is connected to the risk-neutral terminal density through the Breeden-Litzenberger relation

$$\partial_{KK} C(K,T) = e^{-rT} f^{\mathbb{Q}}_{S_T}(K)$$

under regularity and constant rates. Calendar constraints require appropriately discounted or forward-normalized prices to be consistent across maturity [15, 14].

6.5 Local volatility

Local-volatility model

Under the risk-neutral measure, a local-volatility model has dynamics

$$\frac{\mathrm{d}S_t}{S_t} = (r-q)\,\mathrm{d}t + \sigma_{\mathrm{loc}}(t,S_t)\,\mathrm{d}W^{\mathbb{Q}}_t,$$

where volatility is a deterministic function of current time and current spot.

The model remains one-dimensional and Markov in S_t, but its transition law can generate non-lognormal terminal distributions and fit a full smooth surface of European vanilla option prices.

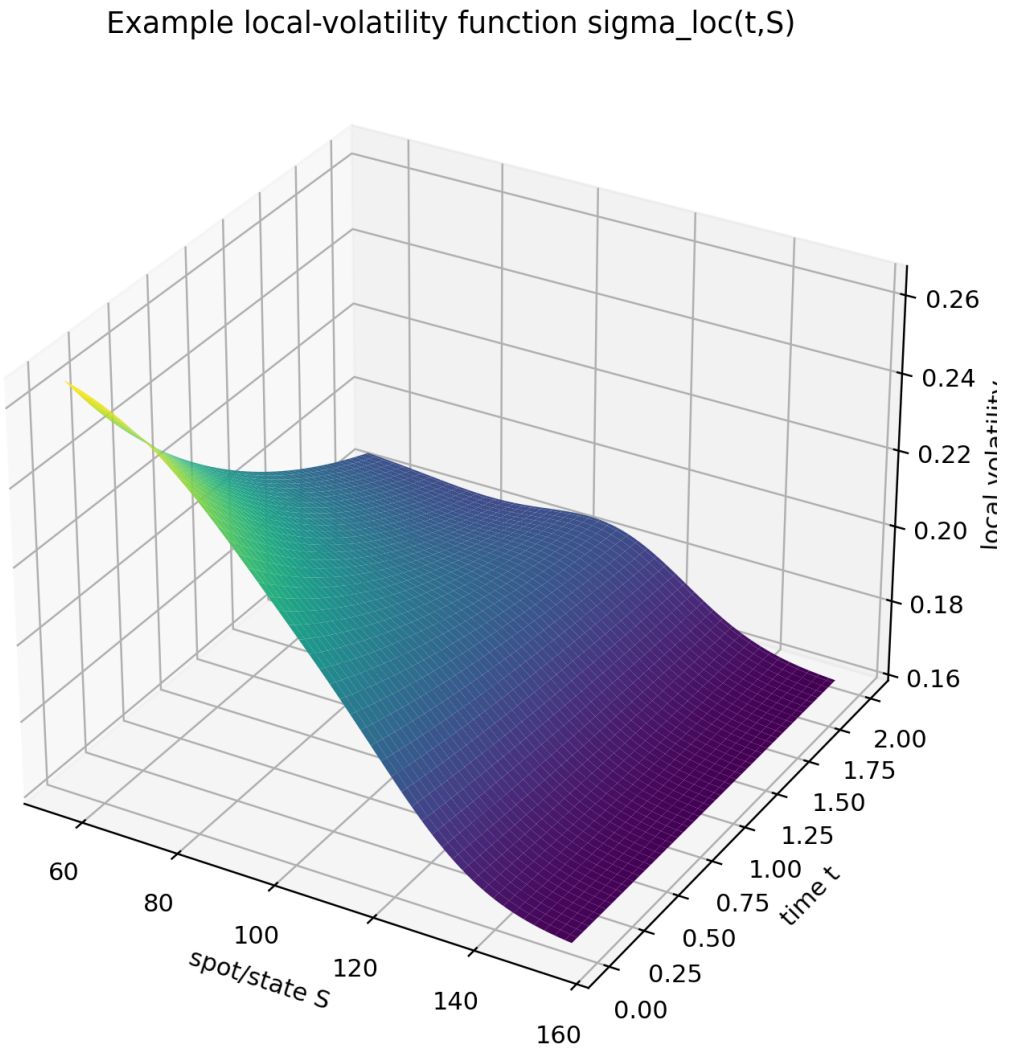


Figure 6.3: An example local-volatility function. At each point, volatility is fixed by current time and state.

6.6 Dupire's forward equation

Treat a call price as a function of strike and maturity, $C(K, T)$. Under deterministic constant r and q, the forward equation is

$$\partial_T C = \frac{1}{2}\sigma_{\text{loc}}^2(T, K)K^2\partial_{KK}C - (r - q)K\partial_K C - qC.$$

Solving for local variance gives Dupire's formula:

$$\boxed{\sigma_{\text{loc}}^2(T, K) = \frac{\partial_T C + (r - q)K\partial_K C + qC}{\frac{1}{2}K^2\partial_{KK}C}.}$$

The state argument appears as K because the formula is written in strike space. Once recovered, the same numerical function is evaluated as $\sigma_{\text{loc}}(t, S_t)$ in the SDE [12, 13].

6.7 What is implied local volatility?

Implied local volatility is the local-volatility function inferred from a surface of vanilla option prices. It is analogous to implied volatility only in the broad sense of inversion:

- implied volatility inverts one Black-Scholes price at one (K, T);
- implied local volatility differentiates an entire option-price surface and inverts a forward equation.

The second operation is much more fragile. Dupire uses a time derivative, a strike derivative, and especially a second strike derivative. Differentiation amplifies quote noise. A practical pipeline therefore requires:

1. cleaning and converting quotes to consistent forwards and discount factors;
2. enforcing or checking static no-arbitrage constraints;
3. fitting a smooth price, total-variance, or density representation;
4. differentiating the smooth representation;
5. flooring or regularizing unstable regions only with explicit diagnostics;
6. repricing vanilla options and testing out-of-sample stability.

Experiment: Dupire sanity check

A dense analytic Black-Scholes call surface with true volatility 24% was differentiated numerically. On an interior region away from unstable boundaries, the median recovered local volatility was 0.2400022. The median absolute error was 3.82×10^{-6} and the 95th percentile error was 3.62×10^{-5}. This clean result is a best-case test; market data are much noisier.

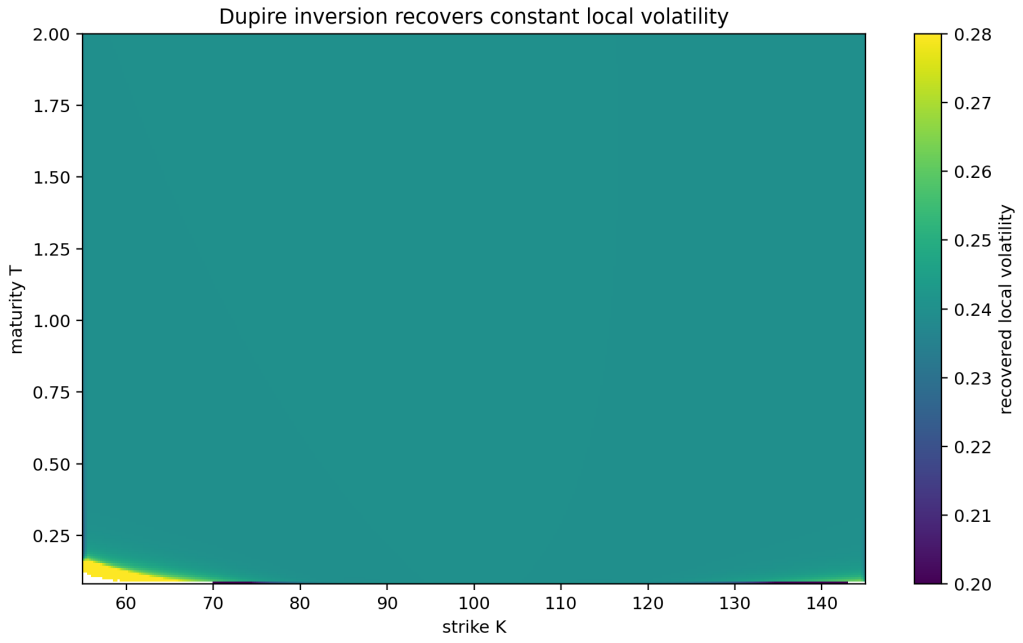


Figure 6.4: Dupire inversion of a constant-volatility Black-Scholes surface. Boundary and very-low-density regions are numerically less stable.

6.8 Local volatility versus stochastic volatility

A stochastic-volatility model introduces a random variance state, for example

$$\begin{aligned} \frac{\mathrm{d}S_t}{S_t} &= (r-q)\,\mathrm{d}t + \sqrt{V_t}\,\mathrm{d}W_t^S, \\ \mathrm{d}V_t &= \kappa(\theta - V_t)\,\mathrm{d}t + \xi\sqrt{V_t}\,\mathrm{d}W_t^V, \\ \mathrm{d}\langle W^S, W^V\rangle_t &= \rho\,\mathrm{d}t. \end{aligned}$$

Here future volatility remains random even after conditioning on current spot. Local volatility instead assigns a single instantaneous value to each (t, S).

Under regularity, a one-dimensional local-volatility diffusion can mimic the one-time marginal distributions of a stochastic-volatility model. In a common form,

$$\sigma_{\mathrm{loc}}^2(t,s) = \mathbb{E}^{\mathbb{Q}}[V_t \mid S_t = s].$$

This projection result explains how local volatility can reproduce European vanilla prices while disagreeing on forward-smile dynamics and path-dependent options [16].

6.9 Identifiability and stability of volatility surfaces

A smooth continuum of exact call prices can identify a risk-neutral marginal density at each maturity and, under Dupire assumptions, a local-volatility surface. Real data are finite, noisy, asynchronous, and concentrated around liquid strikes. Consequently:

- interpolation choices affect derivatives;
- wings and short maturities are weakly constrained;
- several stochastic-volatility or jump models can fit similar vanilla surfaces;
- local-volatility values in data-poor regions may be artifacts of regularization.

Uniqueness inside a mathematical model does not imply stable recovery from finite observations. This distinction between identifiability and conditioning will be formalized in Chapter 9.

A surface can be observed only through instruments

The market does not hand us a continuous function $C(K,T)$. It gives a changing set of bid/ask quotes. A continuous surface is a model fitted between and beyond those points. Its derivatives inherit both market noise and modeling choices.

Chapter summary

- Instantaneous volatility scales standard deviation per square-root time.
- Realized volatility is estimated from returns; implied volatility re-parameterizes an option price.
- The smile is a surface of implied parameters, not direct proof of one unique dynamic model.
- Local volatility is a deterministic function of time and spot under the pricing measure.
- Dupire's formula recovers local variance from derivatives of a smooth arbitrage-consistent option surface.
- Local volatility can match European marginals yet misprice path-dependent dynamics relative to other models.

Chapter 7

Sensitivities, computational flow, and automatic differentiation

7.1 What is a sensitivity?

A sensitivity measures how an output changes when an input or parameter is perturbed. If a scalar price is $V(\theta)$, the first-order sensitivity to parameter θ_j is

$$\frac{\partial V}{\partial \theta_j}.$$

For a small perturbation $\delta\theta$,

$$V(\theta + \delta\theta) \approx V(\theta) + \nabla V(\theta)^\top \delta\theta.$$

The approximation is local. Large shocks require curvature terms or full repricing.

In finance, sensitivities support hedging, risk limits, calibration, scenario analysis, reserve calculations, and numerical diagnostics. A sensitivity can be with respect to a market input, model parameter, state variable, or even an entire curve or surface.

7.2 The principal Greeks

For a value $V(t, S, \sigma, r, \ldots)$:

$$\Delta = \frac{\partial V}{\partial S}, \qquad \Gamma = \frac{\partial^2 V}{\partial S^2},$$
$$\nu = \frac{\partial V}{\partial \sigma}, \qquad \Theta = \frac{\partial V}{\partial t},$$
$$\rho = \frac{\partial V}{\partial r}.$$

Vega is conventionally denoted by ν even though it is not a Greek letter named "Vega" in classical Greek.

For a Black-Scholes call with $\tau = T - t$,

$$\Delta = e^{-q\tau} N(d_1), \qquad \Gamma = \frac{e^{-q\tau}\phi(d_1)}{S\sigma\sqrt{\tau}}, \qquad \nu = S e^{-q\tau}\phi(d_1)\sqrt{\tau}.$$

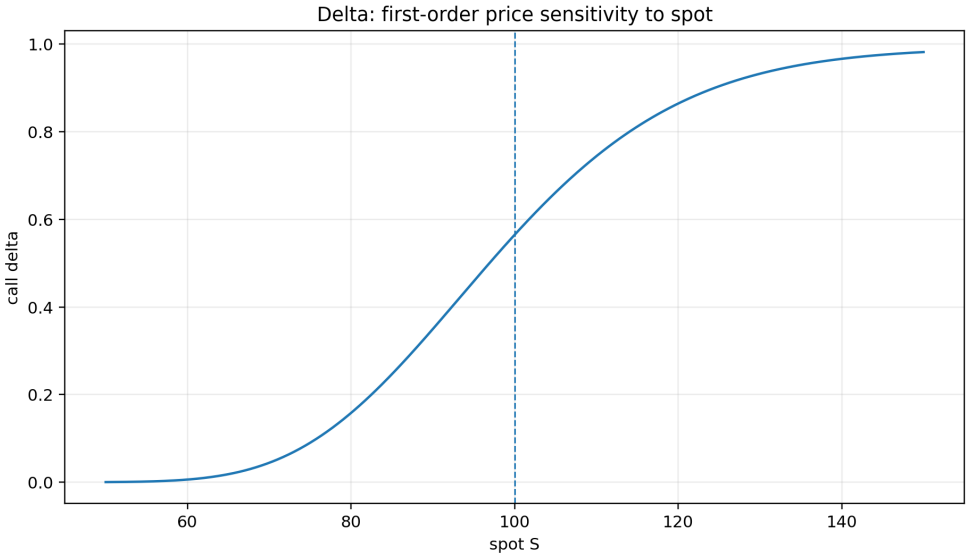


Figure 7.1: Call Delta transitions from near zero to near the discounted share exposure as spot moves through the strike.

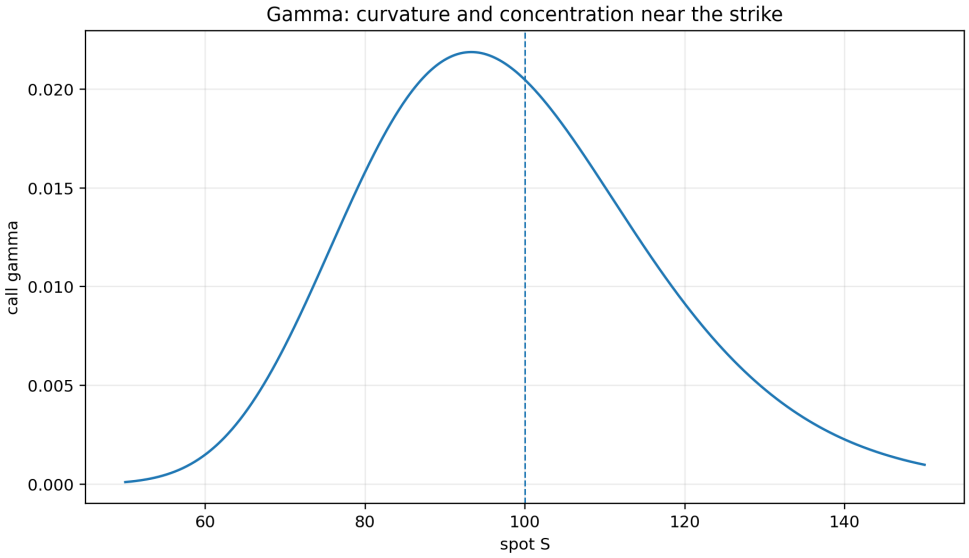


Figure 7.2: Call Gamma is concentrated around the strike and grows sharper as maturity decreases.

7.2.1 Interpretation of Delta and Gamma

A delta hedge offsets the first-order effect of a small spot move. If a portfolio has Delta zero, its leading spot risk is approximately

$$\frac{1}{2}\Gamma(\delta S)^2.$$

Gamma therefore measures curvature and the instability of Delta. A high-Gamma position needs more frequent rebalancing under a continuous-hedging approximation and is more exposed to discrete jumps.

7.3 Finite differences

The central finite-difference approximation is

$$\frac{\partial V}{\partial \theta} \approx \frac{V(\theta + h) - V(\theta - h)}{2h}.$$

It has truncation error $O(h^2)$ for a smooth function, but floating-point cancellation grows when h is too small. The total error is often U-shaped as a function of h.

Second derivatives can be approximated by

$$\frac{\partial^2 V}{\partial \theta^2} \approx \frac{V(\theta + h) - 2V(\theta) + V(\theta - h)}{h^2}.$$

Finite differences are simple and model-agnostic, but they may require one or more full repricings per input, and Monte Carlo noise can dominate small differences.

Experiment: Finite-difference Delta

For an at-the-money Black-Scholes call with $S = K = 100$, $T = 1$, $r = 3\%$, $q = 1\%$, and $\sigma = 22\%$, the central finite-difference Delta was 0.5738478975. The analytic Delta was 0.5738479031. The difference is about 5.6×10^{-9}.

7.4 Computational flow

A computational flow or computational graph expresses an output as a directed acyclic graph of elementary operations. For the Black-Scholes formula, inputs flow through logarithms, products, square roots, d_1, d_2, normal cumulative functions, discount factors, and finally the price.

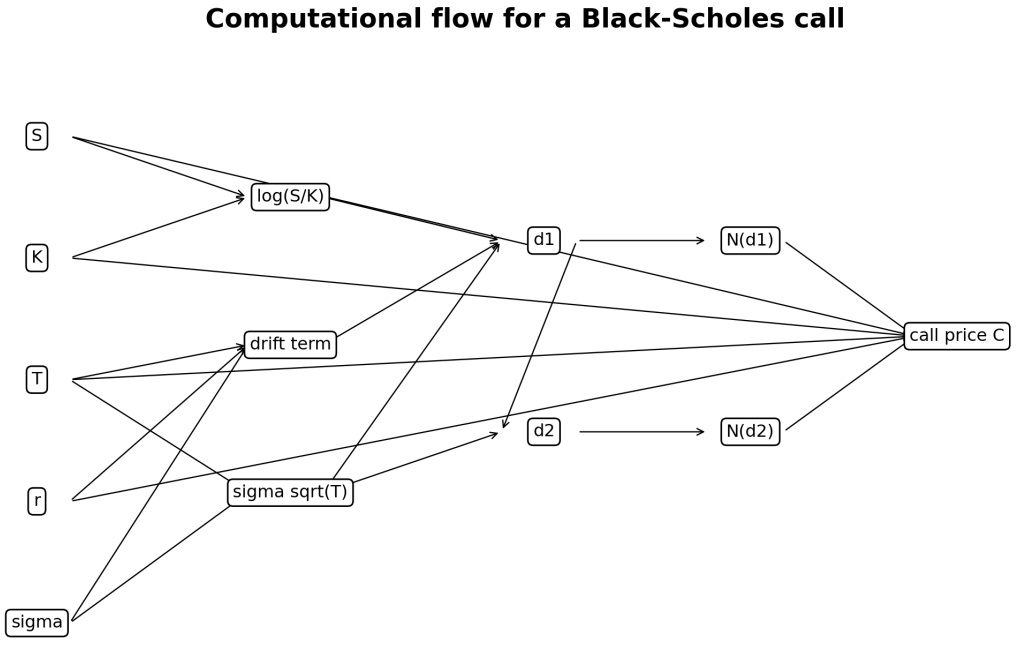


Figure 7.3: A computational graph exposes where the chain rule must propagate derivatives.

If a node is $z = f(x, y)$, then a perturbation obeys

$$\mathrm{d}z = f_x \,\mathrm{d}x + f_y \,\mathrm{d}y.$$

Automatic differentiation applies this chain rule exactly to the sequence of floating-point operations. It is neither symbolic differentiation nor finite differencing.

7.4.1 Forward mode

Forward mode propagates directional derivatives with the primal values. For a map $F : \mathbb{R}^d \to \mathbb{R}^m$, one pass computes

$$J_F(\theta)v$$

for a chosen direction v. It is efficient when the number of input directions is small.

7.4.2 Reverse mode

Reverse mode first evaluates the graph, then propagates adjoints backward. One reverse pass for a scalar output computes

$$\nabla_\theta V.$$

It is efficient when there are many inputs and few outputs. Backpropagation in neural networks and adjoint algorithmic differentiation in finance are reverse-mode methods [17].

7.5 Monte Carlo sensitivities

For

$$V(\theta) = \mathbb{E}[g(X^\theta)],$$

three major approaches are used.

7.5.1 Pathwise differentiation

If the simulated path and payoff are differentiable enough,

$$\frac{\partial V}{\partial \theta} = \mathbb{E}\left[\frac{\partial}{\partial \theta} g(X^\theta)\right].$$

Common random numbers reduce variance when comparing nearby parameters. Discontinuous payoffs such as digitals and barriers require care because the pathwise derivative may not exist in the naive form.

7.5.2 Likelihood-ratio method

If the density $p_\theta(x)$ is differentiable,

$$\frac{\partial}{\partial \theta}\mathbb{E}_\theta[g(X)] = \mathbb{E}_\theta\left[g(X)\frac{\partial}{\partial \theta}\log p_\theta(X)\right].$$

This method differentiates the probability weight rather than the payoff. It can handle discontinuous payoffs but may have high variance.

7.5.3 Adjoint differentiation

For a large simulation and many parameters, reverse-mode differentiation through the path generator and payoff can compute a gradient at a cost that is a modest multiple of one valuation, provided memory and checkpointing are managed. Nonsmooth branching, stopping rules, and calibration solvers need specialized treatment.

7.6 Sensitivities to curves and surfaces

A price may depend on an entire yield curve $r(u)$ or local-volatility surface $\sigma(t, s)$. The derivative is then a functional derivative. A small perturbation can be written

$$\delta V \approx \int \frac{\delta V}{\delta r(u)} \delta r(u)\, \mathrm{d}u + \iint \frac{\delta V}{\delta \sigma(t,s)} \delta\sigma(t,s)\, \mathrm{d}t\, \mathrm{d}s.$$

In practice, curves are parameterized by knots, basis functions, or market instruments, converting functional sensitivities into a finite vector. The result depends on the bump basis: a parallel shift, a single-node bump, and a localized spline bump are different risk scenarios.

7.7 The Jacobian of a price map

Suppose d parameters produce m instrument prices:

$$\mathcal{P} : \mathbb{R}^d \to \mathbb{R}^m, \qquad \theta \mapsto (P_1(\theta), \ldots, P_m(\theta)).$$

The Jacobian is

$$J_{ij}(\theta) = \frac{\partial P_i}{\partial \theta_j}.$$

Each column is the cross-instrument effect of one parameter; each row is the parameter sensitivity of one instrument.

The linearized calibration relation is

$$\delta p \approx J(\theta)\delta\theta.$$

If a data perturbation δp is inverted using a poorly conditioned J, a small quote change can produce a large parameter change. This is why Greeks, identifiability, and numerical stability are the same local differential object viewed from different directions.

7.8 Second-order sensitivities and Hessians

The Hessian $H = \nabla^2 V$ captures curvature in parameter space:

$$V(\theta + \delta\theta) \approx V(\theta) + \nabla V^\top \delta\theta + \frac{1}{2}\delta\theta^\top H \delta\theta.$$

Mixed derivatives quantify interaction, such as Vanna $\partial^2 V/(\partial S \partial\sigma)$ and Volga/Vomma $\partial^2 V/\partial\sigma^2$. For calibration objectives, the Gauss-Newton approximation uses $J^\top W J$, while a full Hessian also includes residual-weighted second derivatives of the price map.

A sensitivity is tied to a coordinate system

Changing from volatility to variance, from strike to log-moneyness, or from raw parameters to normalized parameters changes derivative magnitudes. Sensitivities should be reported with bump size, units, and parameterization.

Chapter summary

- Sensitivities are local derivatives; Greeks are sensitivities to standard market and model inputs.
- Finite differences trade truncation error against floating-point and simulation noise.
- Computational graphs organize the chain rule; reverse mode is efficient for one price and many inputs.
- Pathwise, likelihood-ratio, and adjoint methods address Monte Carlo Greeks differently.
- A Jacobian simultaneously describes calibration information, risk transmission, and local numerical conditioning.

Chapter 8

Numerical methods: Monte Carlo, LSMC, particles, splines, and convolution

8.1 Four numerical questions

A computational method should be evaluated along four axes:

1. **Bias:** does the expected numerical answer differ from the mathematical target?
2. **Variance:** how much does the estimate fluctuate across samples?
3. **Stability:** do small input or roundoff perturbations cause large output changes?
4. **Cost:** how do memory and runtime scale with paths, time steps, state dimension, and parameter count?

A visually smooth result may still be biased; an unbiased estimator may be too noisy to use; a precise calibration may be unstable to a one-tick quote change.

8.2 Monte Carlo revisited

For a discounted payoff Y, the sample mean

$$\widehat{V}_N = \frac{1}{N}\sum_{i=1}^{N} Y_i$$

is unbiased if Y_i are unbiased independent samples. Its variance is $\mathrm{Var}(Y)/N$.

A confidence interval based on a central limit approximation is

$$\widehat{V}_N \pm z_{1-\alpha/2}\frac{s_Y}{\sqrt{N}}.$$

This interval quantifies sampling uncertainty, not time-discretization error, model error, parameter error, or data error.

8.2.1 Discretization bias

For a path-dependent contract, simulate both path values and event detection. A discretely monitored barrier can miss crossings between grid points. Refining the grid changes the contract approximation and the SDE approximation. Brownian-bridge corrections can estimate between-step crossings under compatible models.

8.2.2 Common random numbers

When comparing two parameter sets, reusing the same random numbers makes pathwise differences positively correlated and often dramatically reduces the variance of the price difference. This is especially important for finite-difference Greeks and calibration objectives.

8.3 PDE and finite-difference methods

A Markovian derivative value often solves

$$V_t + \mathcal{L}V - rV = 0$$

with terminal and boundary conditions. A finite-difference method discretizes time and state, replacing derivatives by local stencils.

Explicit schemes are simple but can require restrictive step sizes. Implicit schemes require solving a linear system but are more stable. Crank-Nicolson averages explicit and implicit updates and is second-order in time for smooth solutions, though payoff kinks may require damping.

The curse of dimensionality is severe: if each of d state dimensions has n grid points, the grid has n^d cells. Monte Carlo is often preferable in high dimensions because its nominal $N^{-1/2}$ rate does not directly deteriorate with dimension, although payoff variance and path construction can.

8.4 Least-squares Monte Carlo

American exercise requires comparing immediate payoff with continuation value. At exercise date t_j, define

$$C_j(x) = \mathbb{E}^{\mathbb{Q}}\left[e^{-r\Delta t} V_{j+1}(S_{t_{j+1}}) \mid S_{t_j} = x\right].$$

The Longstaff-Schwartz method estimates C_j by least-squares regression on simulated in-the-money paths [11].

8.4.1 Algorithm

1. Simulate risk-neutral paths on all exercise dates.
2. Set terminal cash flows equal to the payoff at maturity.
3. Work backward through exercise dates.
4. On in-the-money paths, regress discounted future realized cash flows on basis functions of the current state.
5. Exercise when immediate payoff exceeds the fitted continuation value.
6. Discount the selected cash flows to time zero and average.

With basis vector $\phi(x) = (1, x, x^2, \ldots)$, the regression solves

$$\widehat{\beta}_j = \arg\min_{\beta} \sum_{i \in \mathcal{I}_j} \left(Y_i - \phi(S_{t_j}^{(i)})^\top \beta\right)^2.$$

Experiment: American put by LSMC

For $S_0 = K = 100$, $r = 5\%$, $\sigma = 25\%$, one year, 50 exercise dates, 110,000 paths, and a cubic polynomial continuation basis, the estimated American-put value was 7.9355. The European Black-Scholes put was 7.4589, giving an estimated early-exercise premium of 0.4765.

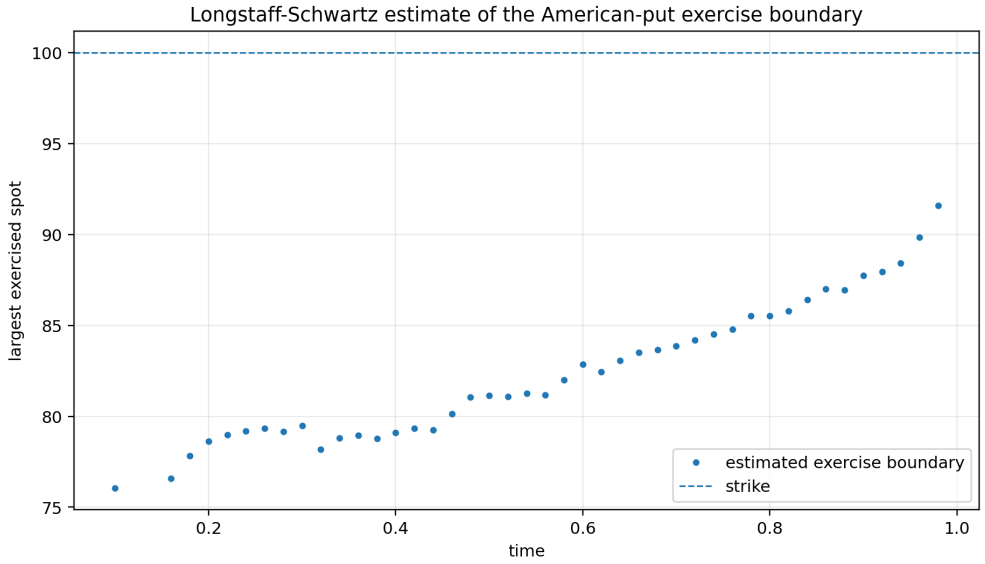


Figure 8.1: A simulation estimate of the American-put exercise boundary. Regression noise and finite sampling can make the raw boundary irregular.

8.4.2 Bias and validation in LSMC

The same simulated future cash flows are used to fit and exercise, which creates estimation subtleties. Basis choice, state variables, path count, and exercise grid all matter. Useful checks include:

- compare with a binomial tree or PDE in one dimension;
- use independent paths to evaluate a learned stopping policy;
- increase basis richness and path count separately;
- confirm monotonicity and obvious lower/upper bounds;
- inspect exercise regions rather than only the scalar price.

8.5 Particle methods

Particle systems can estimate distributions, filter hidden states, or approximate nonlinear dynamics. In a particle filter:

1. propagate particles through the state dynamics;
2. weight them by observation likelihood;
3. normalize the weights;
4. resample when effective sample size becomes too small.

Weight degeneracy is a central problem: after many observations, only a few particles may carry meaningful weight. Resampling reduces weight concentration but adds sampling noise and can reduce diversity. Proposal design and rejuvenation steps are therefore important [19].

8.6 Polynomials and splines

A polynomial approximation has global basis functions:

$$f(x) \approx \sum_{k=0}^{p} a_k x^k.$$

Polynomials are simple and differentiable but high degrees can be ill-conditioned, especially in the monomial basis. Orthogonal polynomials such as Chebyshev or Legendre polynomials improve conditioning on standardized domains.

A spline is piecewise polynomial with continuity constraints at knots. A cubic spline is usually twice continuously differentiable. Splines offer local control: changing data near one knot has less global impact than changing a high-degree global polynomial.

In volatility-surface work, splines can interpolate quotes but must be constrained or parameterized carefully to avoid static arbitrage. Smoothness alone is not enough; convexity and monotonicity conditions have financial meaning.

8.7 Convolution

For functions f and k, the convolution is

$$(f * k)(t) = \int_{-\infty}^{\infty} f(t-u)k(u)\,\mathrm{d}u.$$

In discrete time,

$$y_n = \sum_j k_j x_{n-j}.$$

The kernel k determines how neighboring observations contribute. A moving average is convolution with a flat finite kernel. A Gaussian kernel provides smooth weighting. Derivative kernels emphasize changes rather than levels.

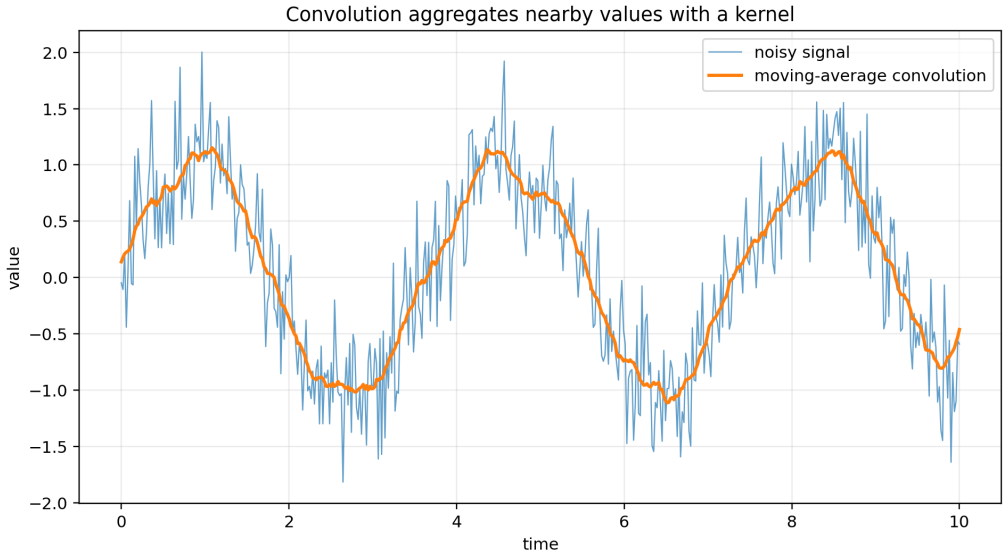


Figure 8.2: A moving-average convolution reduces high-frequency variation but also blurs sharp features and creates boundary effects.

Convolution is a smoothing operator, which often means its inverse is unstable: high-frequency information is attenuated, so deconvolution amplifies noise. This is a prototype of an ill-posed inverse problem.

8.8 Interpolation versus regression

Interpolation passes exactly through supplied points. Regression balances fit against noise or model complexity. In noisy markets, exact interpolation can overfit bid/ask fluctuations and produce unstable derivatives. A penalized fit solves, schematically,

$$\min_f \sum_i w_i (f(x_i) - y_i)^2 + \lambda \mathcal{R}(f),$$

where $\mathcal{R}$ penalizes roughness, curvature, or parameter size.

The regularization parameter λ trades fit against stability. Choosing it is part of the model, even when software presents a default.

8.9 Numerical linear algebra matters

Least squares should generally be solved with QR or singular-value decomposition rather than normal equations. The normal equations

$$X^\top X \beta = X^\top y$$

square the condition number. SVD also reveals rank deficiency and permits truncated or ridge-regularized solutions.

For a matrix $A = U\Sigma V^\top$, a small singular value means some input direction produces little output change. Dividing by that singular value during inversion amplifies noise. Numerical stability therefore links directly to identifiability.

8.10 A practical error budget

For a reported price or parameter, separate:

$$\begin{aligned}\text{total discrepancy} \approx\ & \text{sampling error} + \text{time/state discretization} \\ & + \text{solver tolerance} + \text{input uncertainty} + \text{model error}.\end{aligned}$$

The terms do not literally add independently in every setting, but the decomposition prevents a tiny Monte Carlo standard error from being mistaken for total reliability.

Chapter summary

- Numerical quality requires separate analysis of bias, variance, stability, and cost.
- Monte Carlo confidence intervals do not include discretization or model error.
- LSMC converts American exercise into backward conditional-expectation regression.
- Splines give local smooth approximations; high-degree monomial polynomials can be unstable.
- Convolution smooths signals but can destroy information needed by an inverse problem.
- SVD exposes weakly informed directions and supports stable regularization.

Chapter 9

Price maps, uniqueness, identifiability, rank, manifolds, and gauge freedom

9.1 The forward price map

Let $\theta \in \Theta$ denote model inputs: perhaps a scalar volatility, a vector of stochastic-volatility parameters, spline coefficients, or a discretized local-volatility surface. Let observed instruments be indexed by $i = 1, \dots, m$. The forward or price map is

$$\mathcal{P} : \Theta \to \mathbb{R}^m, \qquad \mathcal{P}(\theta) = (P_1(\theta), \dots, P_m(\theta)).$$

Pricing evaluates $\mathcal{P}$. Calibration tries to solve

$$\mathcal{P}(\theta) = p^{\mathrm{mkt}}$$

or, with noise and weights,

$$\widehat{\theta} = \arg\min_{\theta \in \Theta} \left\| W^{1/2}(\mathcal{P}(\theta) - p^{\mathrm{mkt}}) \right\|^2 + \lambda R(\theta).$$

9.2 Existence, uniqueness, and identifiability

These terms answer different questions.

Existence

A solution exists if at least one admissible θ produces the target observations or minimizes the stated objective.

Uniqueness

A solution is unique if exactly one admissible θ solves the mathematical problem under the chosen model, domain, constraints, and data.

Identifiability

A parameter is globally identifiable if

$$\mathcal{P}(\theta_1) = \mathcal{P}(\theta_2) \quad \Longrightarrow \quad \theta_1 = \theta_2.$$

It is locally identifiable at θ_0 if this implication holds within some neighborhood of θ_0.

A unique optimizer of a regularized objective need not mean the data identify the parameter. The penalty may select one member from many observationally equivalent solutions. Conversely, a globally identifiable noiseless model may be practically unrecoverable if the inverse is extremely unstable.

9.3 Injectivity and local injectivity

A map is injective if different inputs always give different outputs. It is locally injective near θ_0 if this holds only inside a neighborhood.

For a differentiable map $\mathcal{P} : \mathbb{R}^d \to \mathbb{R}^m$ with $m \geq d$, if the Jacobian has full column rank d at θ_0, a nonzero $d \times d$ minor remains nonzero nearby. The constant-rank theorem then gives local coordinates in which the map looks like

$$(x_1, \ldots, x_d) \mapsto (x_1, \ldots, x_d, 0, \ldots, 0),$$

so the map is locally one-to-one onto its image. When $m = d$ and $\det J(\theta_0) \neq 0$, the inverse function theorem gives a local differentiable inverse [20].

Full rank is a local statement. A map can be locally injective everywhere yet not globally injective, as with a periodic parameterization over a domain containing multiple periods.

9.4 Full rank and what it means

For an $m \times d$ Jacobian J:

- full column rank means $\operatorname{rank} J = d$ and requires $m \geq d$;
- full row rank means $\operatorname{rank} J = m$ and requires $d \geq m$;
- a square matrix is full rank when its determinant is nonzero.

Full column rank means no nonzero infinitesimal parameter perturbation is completely invisible:

$$J\delta\theta = 0 \quad \Longrightarrow \quad \delta\theta = 0.$$

Equivalently, $\ker J = \{0\}$.

9.5 Kernel, invisible directions, and "completeness"

In inverse problems, the kernel of the derivative contains perturbations that do not change observations to first order. If a function-valued parameter h is mapped by a linearized pricing operator A to price changes, then

$$Ah = 0$$

means the instrument set cannot detect perturbation h.

The phrase "completeness of the kernel" is not standard without context. Two precise nearby ideas are:

1. **Trivial nullspace:** $\ker A = \{0\}$, giving injectivity of a linear forward operator.
2. **Complete feature or instrument family:** the span of measurement functionals is dense enough to distinguish every admissible object.

A finite instrument set cannot identify an unrestricted infinite-dimensional surface without structural assumptions or regularization.

9.6 Singular values and stability

Let

$$J = U \operatorname{diag}(s_1, \ldots, s_r) V^\top, \qquad s_1 \geq \cdots \geq s_r > 0.$$

The right singular vector v_i is a parameter direction, and s_i is the magnitude of the resulting first-order price movement. A small s_i means weak information.

For a locally square invertible map, perturbations satisfy approximately

$$\delta\theta \approx J^{-1}\delta p.$$

The amplification is controlled by

$$\left\| J^{-1} \right\|_2 = \frac{1}{s_{\min}}.$$

The condition number

$$\kappa_2(J) = \frac{s_{\max}}{s_{\min}}$$

measures relative anisotropy. A full-rank Jacobian can still be badly conditioned if $s_{\min}$ is positive but tiny.

Experiment: Two prices identify volatility and rate locally

Two one-year Black-Scholes calls with strikes 90 and 115 were treated as observations for unknown (σ, r) around $(0.23, 0.028)$. Their equal-price contours cross at the true pair. The local Jacobian was approximately

$$\begin{pmatrix} 31.3382 & 59.4203 \\ 37.2421 & 30.6339 \end{pmatrix},$$

with singular values 81.2430 and 15.4219, giving condition number 5.2680.

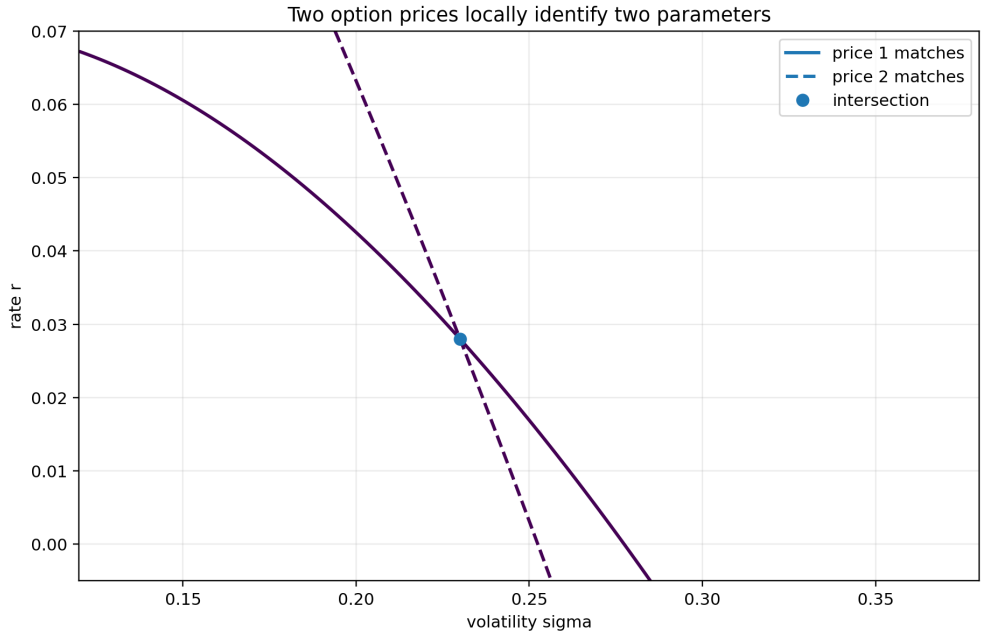


Figure 9.1: One option price defines a curve of matching parameters; two independent prices can isolate a local intersection.

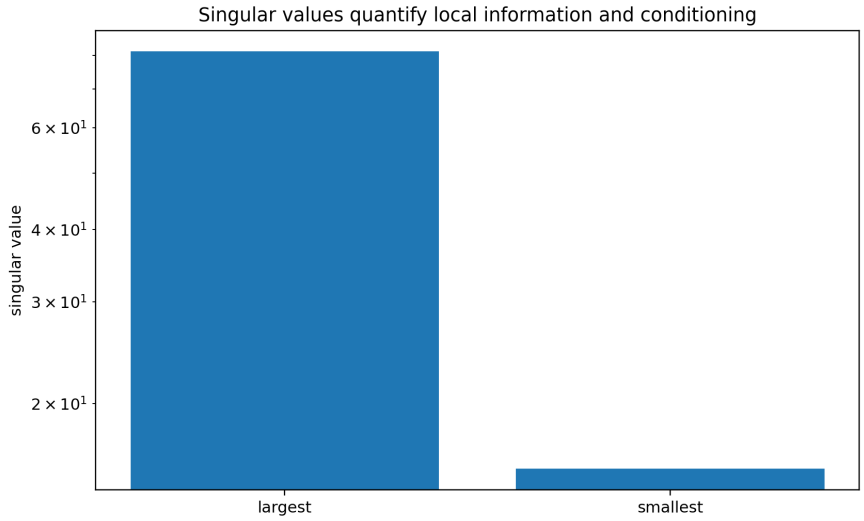


Figure 9.2: Singular values summarize the strongest and weakest locally observable parameter combinations.

9.7 Stability of Jacobians

A calibration may be stable at one point and unstable nearby. Questions to ask include:

- Does the smallest singular value remain bounded away from zero across plausible parameters?
- Do finite-difference Jacobians change materially with bump size?
- Are columns badly scaled because parameters use different units?
- Does the instrument set lose sensitivity in certain regimes, such as very short maturities or

saturated options?

- Are numerical solver tolerances smaller than quote noise, or merely over-resolving the model?

A normalized Jacobian can be more informative:

$$\widetilde{J}_{ij} = \frac{\theta_j}{P_i}\frac{\partial P_i}{\partial \theta_j},$$

when prices and parameters are nonzero. This compares percentage effects, though the chosen normalization still reflects a coordinate system.

9.8 Manifolds of prices and solutions

If $\mathcal{P} : \mathbb{R}^d \to \mathbb{R}^m$ has constant rank r, its image is locally an r-dimensional immersed manifold. When $m > d$ and rank is d, model prices occupy a thin d-dimensional surface in the m-dimensional quote space. Market observations off that surface cannot be fit exactly.

For a level set

$$\mathcal{M}_y = \{\theta : \mathcal{P}(\theta) = y\},$$

if J has full row rank m with $d > m$, the regular level-set theorem gives local dimension $d - m$. Thus underdetermined calibration solutions form a manifold rather than an isolated point.

The tangent space of a level set is

$$T_\theta \mathcal{M}_y = \ker J(\theta).$$

This equation unifies geometry and identifiability: null directions are tangent directions along which the observations remain unchanged to first order.

9.9 Gauge freedom

Gauge freedom

A gauge freedom is a transformation of internal parameters or representation that leaves all observable quantities unchanged. Parameters related by such transformations represent the same observable model state.

A simple example is

$$y = abx.$$

Observations identify only $c = ab$. For any nonzero λ,

$$(a, b) \mapsto (\lambda a, b/\lambda)$$

leaves the model output unchanged. The curve $ab = c$ is a gauge orbit.

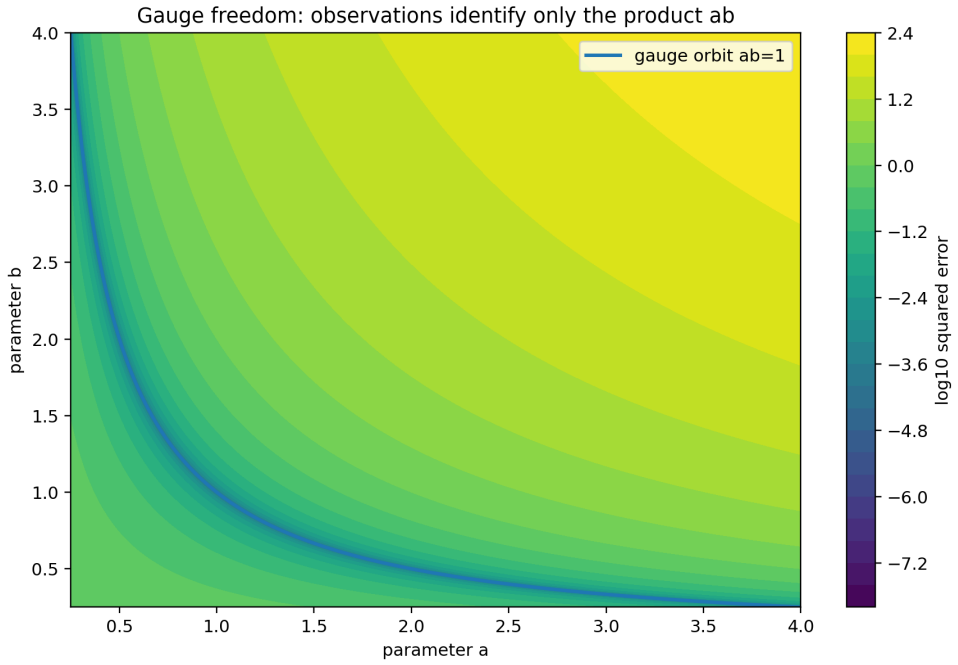


Figure 9.3: The objective is flat along the hyperbola $ab = 1$: infinitely many parameter pairs are observationally equivalent.

Gauge freedoms appear in factor models, neural-network rescalings, matrix factorizations, latent-state coordinate changes, and over-parameterized volatility representations. They should be removed by constraints or quotiented out conceptually rather than mistaken for data uncertainty.

9.10 Global non-uniqueness despite local rank

Even with full-rank Jacobian at every relevant point, a map may wrap around and produce the same output at distant parameter values. Symmetries and label switching are common examples. Global identifiability therefore needs global arguments: monotonicity, domain restrictions, convexity, analyticity, or a proof that no symmetry remains.

9.11 Ill-posed functional inverse problems

A map from a function such as $\sigma(t, s)$ to a finite or smoothed set of option prices often suppresses high-frequency perturbations. Mathematically, linearized pricing operators can resemble compact integral operators whose singular values decay toward zero. Even if the exact continuum map is injective, inversion of noisy finite data is unstable.

Regularization replaces exact inversion with a controlled trade-off:

$$\widehat{\theta}_\lambda = \arg\min_\theta \left\| W^{1/2}(\mathcal{P}(\theta) - p) \right\|^2 + \lambda \left\| L\theta \right\|^2 .$$

L may penalize level, slope, curvature, time roughness, or deviation from a prior. The regularized solution is unique under conditions, but its shape depends on L and λ.

9.12 Identifiability versus estimability

- **Structural identifiability**: with perfect infinite data, can the parameter be uniquely recovered?
- **Practical identifiability**: with actual instruments and noise, are the relevant directions sufficiently sensitive?
- **Estimability**: does a chosen estimator recover the parameter with acceptable bias and variance?

These questions should be reported separately.

A tiny calibration error can hide a huge parameter error

When a Jacobian has a small singular value, the objective is nearly flat along the associated parameter direction. Optimization can fit prices extremely well while the corresponding parameter remains uncertain and unstable. Price fit is not a certificate of parameter truth.

Chapter summary

- The price map sends model parameters to a vector or surface of instrument values.
- Uniqueness belongs to a stated mathematical problem; identifiability asks whether observations distinguish parameters.
- Full column rank gives local observability and local injectivity onto the image, but not global uniqueness.
- The Jacobian kernel contains invisible directions; singular values quantify information strength and conditioning.
- Solution sets and model-price sets are naturally described as manifolds.
- Gauge freedom is exact observational equivalence, not merely poor statistical precision.
- Regularization stabilizes inversion but introduces assumptions that must be acknowledged.

Chapter 10

Functional forms, linear and nonlinear models, MLPs, kernels, and signatures

10.1 What is a functional form?

A functional form is the mathematical shape chosen to relate inputs to outputs. Examples include

$$y = \beta_0 + \beta_1 x, \qquad y = ae^{bx}, \qquad y = \sum_{k=0}^{p} \beta_k x^k, \qquad y = f_\theta(x) \text{ from a neural network.}$$

The choice encodes inductive bias: smoothness, monotonicity, locality, symmetry, scaling, or compositional structure.

A flexible functional form reduces approximation bias but can increase estimation variance, optimization difficulty, and extrapolation risk. The relevant question is not merely "can this model fit?" but "what does it assume between and beyond observations?"

10.2 Linear and nonlinear models

The word *linear* can refer to inputs, parameters, or operators.

10.2.1 Linear in inputs

A model $f(x) = \beta_0 + \beta_1 x$ is affine in x. A polynomial is nonlinear in x when degree exceeds one.

10.2.2 Linear in parameters

A model is linear in parameters if

$$f_\beta(x) = \sum_{j=1}^{p} \beta_j \phi_j(x).$$

Polynomial regression is linear in its coefficients even though it is nonlinear in x. Least squares can estimate β by linear algebra.

10.2.3 Nonlinear in parameters

Models such as

$$f_{a,b}(x) = ae^{bx}$$

or neural networks are nonlinear in parameters. Their objectives may be nonconvex and can contain symmetries, multiple local minima, or flat directions.

10.2.4 Linear operators

An operator A is linear if

$$A(af + bg) = aAf + bAg.$$

Expectation, integration, convolution, and the pricing map from a terminal payoff to its discounted expectation under a fixed model are linear in the payoff. Calibration from model parameters to prices is generally nonlinear.

10.3 Polynomials and splines as function approximators

A polynomial basis is global. Changing one coefficient changes the function everywhere. High-degree monomials can be numerically ill-conditioned and may oscillate near boundaries. Orthogonal polynomial bases and standardized inputs help.

A spline uses low-degree polynomial pieces joined at knots. Cubic splines commonly impose continuity through the second derivative. B-splines provide a stable local basis, and smoothing splines balance residual fit against integrated curvature:

$$\min_f \sum_i (y_i - f(x_i))^2 + \lambda \int (f''(x))^2 \, \mathrm{d}x.$$

The penalty is a concrete expression of a smoothness prior [26, 27].

10.4 Multilayer perceptrons

A feedforward multilayer perceptron (MLP) composes affine maps and nonlinear activations. A one-hidden-layer scalar-output network is

$$f_\theta(x) = \sum_{j=1}^{m} a_j \rho(w_j^\top x + b_j) + c.$$

Deeper networks repeat this construction:

$$h^{(\ell+1)} = \rho(W^{(\ell)} h^{(\ell)} + b^{(\ell)}).$$

Common activations include sigmoid, hyperbolic tangent, ReLU, and smooth variants. Nonlinearity is essential: a composition of purely linear layers collapses to one linear map.

10.4.1 Universal approximation theorem

Classical universal approximation results show that, under conditions on the activation, finite linear combinations of hidden units are dense in continuous functions on compact subsets of $\mathbb{R}^d$ [28, 29]. More modern versions cover broad non-polynomial activations.

What universal approximation does not say

The theorem does not guarantee a small network, an efficient training algorithm, good generalization from finite data, numerical stability, robustness, interpretability, or sensible extrapolation. It is a representational existence result.

10.5 Teacher-student models

A teacher-student model separates data generation from learning.

Teacher-student setup

A teacher with parameter $\theta_\star$ generates examples

$$y_i = f_{\theta_\star}(x_i) + \varepsilon_i.$$

A student model g_θ is trained on the examples and evaluated on new data.

The setup is useful because the ground truth is known. One can study:

- sample complexity and generalization error;
- matched models, where the student class contains the teacher;
- mismatched models, where it does not;
- noise, regularization, over-parameterization, and phase transitions;
- identifiability and symmetries of the parameterization.

In statistical-mechanics analyses of learning, a high-dimensional random training set defines an energy or loss landscape. Order parameters such as overlap between teacher and student weights summarize macroscopic learning behavior [30]. A student can approximate the teacher's function well while its raw parameters differ because of permutations, rescalings, or redundant hidden units.

10.6 Approximation experiment

A smooth target function was sampled at 17 equally spaced points. Three models were fit: a degree-12 polynomial, a cubic spline, and a two-hidden-layer tanh MLP.

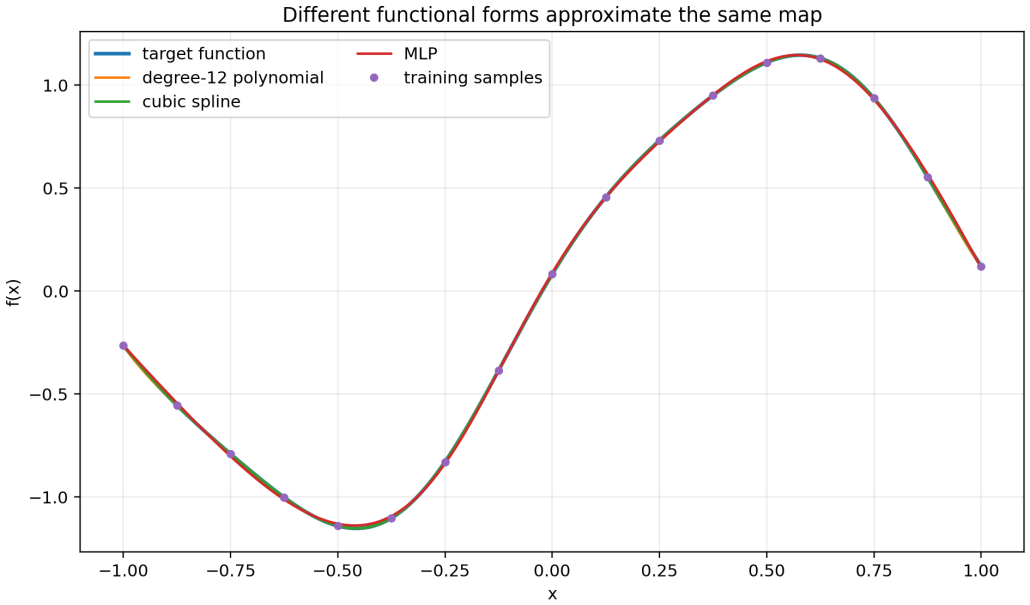


Figure 10.1: Different functional forms interpolate or approximate the same finite observations in different ways.

On a dense evaluation grid, the root mean squared errors were approximately:

$$\text{polynomial } 0.00166, \qquad \text{spline } 0.000215, \qquad \text{MLP } 0.00833.$$

This is not a ranking of model classes. The data were noiseless and favorable to local interpolation. Different sample size, noise, hyperparameters, or target regularity could reverse the result. The experiment demonstrates inductive bias: equal training information does not determine behavior between points.

10.7 Kernel methods

A positive-definite kernel $k(x, x')$ acts as an inner product in a possibly high-dimensional feature space:

$$k(x, x') = \langle \phi(x), \phi(x') \rangle_{\mathcal{H}}.$$

The reproducing property in an RKHS is

$$f(x) = \langle f, k(x, \cdot) \rangle_{\mathcal{H}}.$$

Kernel ridge regression solves

$$\min_{f \in \mathcal{H}} \sum_i (y_i - f(x_i))^2 + \lambda \|f\|_{\mathcal{H}}^2 .$$

By the representer theorem, the solution has finite form

$$f(x) = \sum_{i=1}^{n} \alpha_i k(x_i, x).$$

Thus an infinite-dimensional optimization reduces to n coefficients. Kernel bandwidth controls locality and smoothness; poor bandwidth choice can underfit or overfit [23, 24].

10.8 Convolutional structure

Convolution uses shared local weights:

$$(y * k)_t = \sum_j k_j y_{t-j}.$$

Sharing creates translation equivariance: shifting the input shifts the output. In time-series models, a convolutional filter can detect local motifs such as momentum, reversals, or volatility bursts. Stacked nonlinear convolutions expand the receptive field.

A moving average and a convolutional neural-network layer share the local weighted-sum operation, but a learned layer estimates kernels from data and usually adds channels and nonlinearities.

10.9 Signature series of a path

A path signature converts an ordered path into a sequence of iterated integrals. Let $X : [0, T] \to \mathbb{R}^d$ have suitable regularity. Its signature is

$$\mathcal{S}(X) = \left(1, S^{(1)}, S^{(2)}, \dots\right),$$

where the level-n coordinates are

$$S^{i_1,\dots,i_n}(X) = \int_{0<t_1<\cdots<t_n<T} \mathrm{d}X_{t_1}^{i_1} \cdots \mathrm{d}X_{t_n}^{i_n}.$$

The first level is total displacement:

$$S^i = X_T^i - X_0^i.$$

The second level captures ordered pair interactions and signed area. Higher levels encode increasingly detailed temporal order.

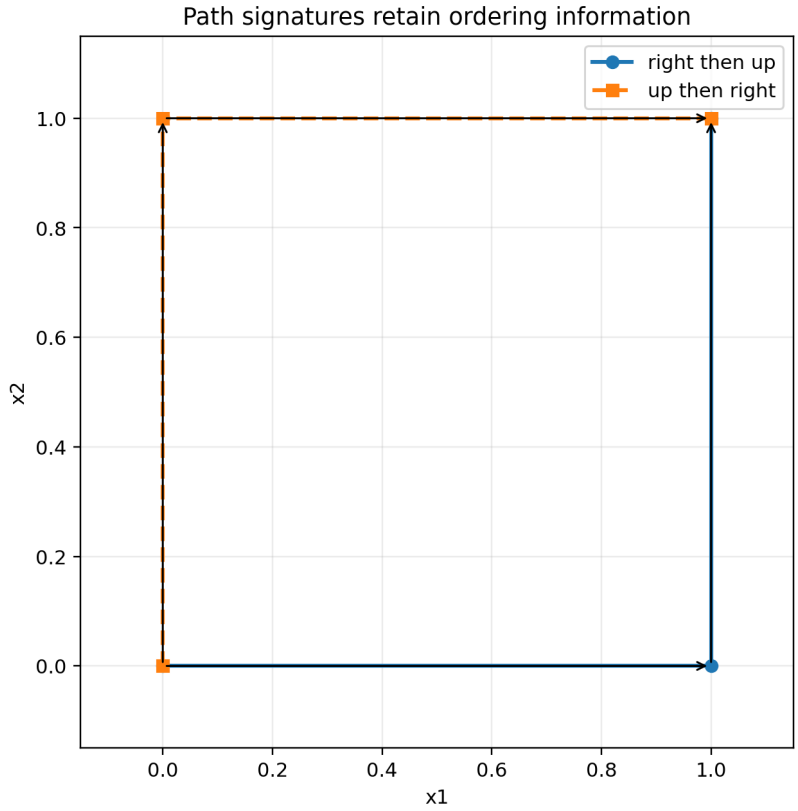


Figure 10.2: Both paths have displacement $(1, 1)$, but their second-level signatures distinguish the order of movement.

For the piecewise-linear path "right then up",

$$S^{1,2} = 1, \qquad S^{2,1} = 0.$$

For "up then right",

$$S^{1,2} = 0, \qquad S^{2,1} = 1.$$

The endpoint alone cannot distinguish them; the signature can.

10.9.1 Chen's identity

If paths X and Y are concatenated, their signatures multiply in the tensor algebra:

$$\mathcal{S}(X * Y) = \mathcal{S}(X) \otimes \mathcal{S}(Y).$$

This compositional identity is valuable for streaming and segmented data.

10.9.2 Uniqueness and universality

Under suitable conditions, the signature determines a path up to tree-like equivalence and monotone time reparameterization. Linear functionals of signatures can approximate broad classes of continuous path functionals on compact sets after appropriate assumptions and augmentation [31, 32].

The raw signature is infinite. In computation it is truncated at level M. The number of coordinates through level M is

$$1 + d + d^2 + \cdots + d^M,$$

which grows rapidly. Log-signatures, sparse selection, low-rank methods, and domain-specific augmentations address this growth.

10.10 Signatures for path-dependent finance

A path-dependent payoff can be approximated as

$$H \approx \sum_{|I| \le M} \beta_I S^I(X).$$

Potential uses include:

- regression for continuation values in American options;
- features for hedging policies;
- classification of market regimes;
- approximation of barriers, averages, and lead-lag effects;
- comparing path laws with signature kernels.

Time and lead-lag channels are often added because the signature is invariant to monotone reparameterization of an unaugmented geometric path. Whether that invariance is desirable depends on the contract.

10.11 Choosing a representation

Representation	Strength	Main risk
Low-degree polynomial	simple, fast, interpretable	global bias or boundary error
Spline	local control and smooth derivatives	knot and constraint choices
Kernel method	convex fit with rich features	cubic scaling and bandwidth sensitivity
MLP	flexible compositional approximation	nonconvex training and extrapolation
Convolution	shared local pattern detection	may miss long-range order without depth/dilation
Signature	principled ordered-path features	feature growth and truncation choice

Representation and identifiability interact

A richer parameterization can fit more functions but may create weakly informed or gauge-equivalent directions. Approximation power should therefore be evaluated together with the singular values of the price or prediction map.

Chapter summary

- A functional form is an inductive bias, not only a formula.
- A model may be nonlinear in inputs but linear in parameters.
- Universal approximation is a representational theorem, not a training or generalization guarantee.
- Teacher-student models expose generalization, mismatch, and parameter symmetries under known ground truth.
- Kernels define implicit feature spaces; convolutions impose local weight sharing.
- Signature series encode ordered path information through iterated integrals and support approximation of path functionals.

Chapter 11

An integrated workflow: from market quotes to path-dependent risk

11.1 The end-to-end problem

Suppose the goal is to use listed vanilla options to calibrate a local-volatility model, price a barrier or American option, and report risk. The full workflow is not a single formula. It is a chain of transformations, each with assumptions and possible instability.

1. Define instruments, conventions, settlement, calendars, and units.
2. Build discount, dividend, and forward curves.
3. Clean option quotes and choose mid, bid/ask, or weighted objectives.
4. Convert quotes to prices or implied volatilities consistently.
5. Fit a smooth static-arbitrage-aware surface.
6. Differentiate the surface to recover local volatility.
7. Simulate or solve the local-volatility pricing equation.
8. Price and validate path-dependent or early-exercise claims.
9. Compute sensitivities through the complete computational flow.
10. Diagnose identifiability, Jacobian conditioning, and model risk.

11.2 Step 1: contract and convention control

A model cannot correct an incorrectly specified contract. Confirm:

- spot or forward underlying;
- strike units and option multiplier;
- exercise style and observation schedule;
- continuous or discrete barrier monitoring;
- dividend, funding, collateral, and discount conventions;
- business-day rules, time zones, and settlement lag.

A one-day maturity mismatch or wrong dividend convention can create an apparent volatility anomaly.

11.3 Step 2: quote representation

For each quote, distinguish:

$$\text{raw premium} \longleftrightarrow \text{forward-normalized price} \longleftrightarrow \text{Black-Scholes implied volatility}.$$

Implied volatility is convenient for comparison but the no-arbitrage constraints live most directly in option prices. A robust surface workflow moves between the two representations carefully.

11.4 Step 3: no-arbitrage and smoothing

At a fixed maturity, verify call prices are decreasing and convex in strike within numerical tolerance. Across maturity, use a representation whose calendar constraints are explicit. Weight liquid near-the-money quotes more heavily only if that matches the use case; a barrier product may be sensitive to wings that are less liquid.

A smooth fit should be judged by:

- weighted pricing error relative to bid/ask;
- static-arbitrage diagnostics;
- stability under removal of a small subset of quotes;
- behavior in extrapolated wings and short maturity;
- derivative smoothness, not only level smoothness.

11.5 Step 4: implied local volatility

Apply Dupire only after smoothing. Record denominator size

$$\frac{1}{2}K^2\partial_{KK}C,$$

because tiny risk-neutral density makes inversion unstable. Flag regions where local variance is determined mainly by extrapolation or regularization.

A surface that reprices its own input quotes is necessary but not sufficient. Check whether small quote perturbations produce large local-volatility changes. This is a direct stability test of the inverse problem.

11.6 Step 5: risk-neutral path generation

Under local volatility,

$$S_{n+1} = S_n + (r-q)S_n\Delta t + \sigma_{\text{loc}}(t_n, S_n)S_n\sqrt{\Delta t}Z_n$$

after a simple Euler discretization. A log-Euler or positivity-preserving scheme may be preferable depending on the surface. Interpolation of σ_{loc} should be stable, bounded where appropriate, and consistent with extrapolation rules.

For a barrier, refine the grid and use Brownian-bridge corrections if supported. For an American option, augment the simulation with backward continuation-value estimation.

11.7 Step 6: pricing validation

Validation should be hierarchical:

1. Recover analytic Black-Scholes prices when local volatility is constant.
2. Reprice the vanilla calibration set.
3. Compare Monte Carlo with a PDE for a one-dimensional European or barrier benchmark.
4. Compare LSMC American values with a tree or PDE benchmark.
5. Test limiting cases: zero volatility, very high barrier, immediate maturity, deep in/out of the money.
6. Verify convergence as paths and time steps increase.

11.8 Step 7: sensitivity flow

Differentiate the entire workflow, not merely the terminal pricing formula. A market-quote sensitivity may pass through:

$$\text{quote} \to \text{surface fit} \to \text{Dupire derivatives} \to \sigma_{\text{loc}} \to \text{path simulation} \to \text{payoff} \to \text{price}.$$

Ignoring the calibration layer produces "model-parameter Greeks" rather than full market Greeks.

For a scalar exotic price and thousands of surface parameters, reverse-mode or adjoint differentiation can be attractive, but nondifferentiable barriers, clipping rules, and regression exercise decisions must be handled explicitly.

11.9 Step 8: identifiability and model risk

Compute the Jacobian of calibration instruments with respect to parameters or basis coefficients. Inspect singular values and right singular vectors. A small singular value identifies a surface deformation that quotes barely detect.

Report at least two types of uncertainty:

- **within-model numerical uncertainty**: Monte Carlo error, grid bias, solver tolerance;
- **between-model uncertainty**: difference among local-volatility, stochastic-volatility, jump, or hybrid models that fit vanilla prices comparably.

For path-dependent products, between-model uncertainty can dominate.

11.10 A compact concept crosswalk

Term	Place in the workflow
Borel set	makes state and path events measurable
Measure	assigns probability or integration weight
Stochastic process	generates uncertain time-indexed states

Path / trajectory	one realized history of the process
SDE	local rule for state evolution
Itô lemma	transforms SDEs and derives pricing dynamics
Martingale	zero conditional drift under chosen measure and information
Strike / moneyness	define contract location relative to spot or forward
Risk-neutral simulation	generates pricing paths under $\mathbb{Q}$
Volatility	instantaneous diffusion scale; meaning depends on model
Implied volatility	one-price Black-Scholes inverse parameter
Local volatility	state- and time-dependent deterministic diffusion scale
Implied local volatility	surface recovered from vanilla prices through Dupire
Sensitivity	derivative of price or calibration output with respect to an input
Computational flow	graph through which values and derivatives propagate
Price map	maps parameters or functions to observable instrument prices
Full rank	no infinitesimal parameter direction is invisible, in the relevant sense
Local injectivity	nearby parameters produce distinct nearby outputs
Kernel	nullspace, transition rule, integral weight, or RKHS similarity, depending on context
Gauge freedom	exact internal transformation that leaves observables unchanged
Manifold	locally Euclidean set of prices, parameters, or equivalent solutions
Particle system	empirical cloud approximating a distribution or interacting law
MLP / spline / polynomial	alternative finite representations of a function
Signature series	ordered iterated-integral representation of a path
LSMC	backward regression method for exercise decisions

11.11 Ten durable principles

1. State the probability measure and filtration.
2. State units and nondimensionalize where possible.
3. Separate payoff specification from dynamic model.
4. Separate physical forecasting from risk-neutral valuation.
5. Treat theorem assumptions as part of the theorem.
6. Distinguish exact uniqueness from stable recoverability.
7. Inspect the kernel and singular values of every inverse map.
8. Validate numerical convergence against known limiting cases.
9. Do not confuse an excellent vanilla fit with correct path dynamics.
10. Report model uncertainty alongside numerical precision.

11.12 A final mental model

The entire subject can be compressed into one diagrammatic equation:

$$\theta \xrightarrow{\text{SDE / law}} \text{paths} \xrightarrow{\text{contract}} H \xrightarrow{\mathbb{E}^{\mathbb{Q}} \text{ and discounting}} P \xrightarrow{\text{calibration inverse}} \widehat{\theta}.$$

The forward arrows are pricing. The backward arrow is inference. Stochastic calculus justifies the path dynamics and transformations. Numerical methods approximate the arrows. Geometry and linear algebra determine whether the backward arrow is unique and stable. Machine-learning and approximation methods choose representations for the functions and paths involved.

The unifying question

At every stage ask: what information enters, what transformation is applied, what information is lost, and how sensitive is the output to perturbations? That question links Borel measurability, martingales, local volatility, automatic differentiation, Jacobian rank, gauge freedom, and universal approximation.

Chapter A

Glossary of the requested terms

Term	Concise definition
Borel set	A member of the sigma-algebra generated by the open sets of a topological space.
Measure	A countably additive assignment of nonnegative size to measurable sets.
Stochastic process	A family of random variables indexed by time or another parameter.
Path / trajectory	The function of time obtained after fixing one outcome of a stochastic process.
Geometric Brownian motion	The positive diffusion $\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t$ with lognormal solution.
Martingale	An adapted integrable process whose conditional future expectation equals its present value.
Martingale theorem	An ambiguous phrase; common intended results are optional stopping, convergence, representation, or the fundamental theorem of asset pricing.
Volatility	Instantaneous standard deviation rate of returns, normally quoted per square-root year.
Local volatility	A deterministic function $\sigma_{\mathrm{loc}}(t, S_t)$ governing instantaneous risk-neutral diffusion.
Implied volatility	The Black-Scholes volatility parameter that reproduces one observed option price.
Implied local volatility	A local-volatility surface inferred from a smooth vanilla option-price surface, commonly through Dupire's formula.
Computational flow	A directed graph of elementary operations used to compute outputs and propagate derivatives.
Sensitivity	A derivative measuring the local response of an output to an input or parameter.
Particle system	A finite collection of random states whose empirical distribution approximates a law, possibly with interactions.
Dimensionless function	A function expressed in unit-free variables and producing a unit-free output.
Strike price	The contractual reference level used to exercise or calculate an option payoff.

Term	Concise definition
Moneyness	A dimensionless comparison of strike with spot or forward, such as K/F or $\log(K/F)$.
Uniqueness	The property that a stated mathematical problem has exactly one admissible solution.
Identifiability	The property that different parameters imply different observable distributions or prices.
Gauge freedom	A transformation of internal parameters that leaves every observable unchanged.
European option	An option exercisable only at maturity.
American option	An option exercisable at any admissible time up to maturity.
Exotic option	An option with non-vanilla payoff, exercise, monitoring, averaging, barrier, reset, or multi-asset features.
Call	An option with payoff increasing above the strike, classically $(S_T - K)^+$.
Put	An option with payoff increasing below the strike, classically $(K - S_T)^+$.
Black-Scholes model	A constant-volatility lognormal diffusion model yielding a replication PDE and closed-form European prices.
Price map	A function mapping model parameters or surfaces to one or more instrument prices.
Full rank	A matrix has the largest rank allowed by its dimensions; for a Jacobian, it means no redundant local direction in the relevant domain or codomain sense.
Local injectivity	Nearby distinct inputs produce distinct outputs.
Manifold	A space that is locally equivalent to Euclidean space, used for model-price and solution sets.
Kernel / nullspace	The set of inputs mapped to zero by a linear map.
Markov kernel	A state-dependent probability measure describing conditional transition probabilities.
Integral kernel	The weighting function inside an integral operator.
Machine-learning kernel	A positive-definite similarity corresponding to an inner product in a feature space.
Complete measure space	A measure space containing every subset of every null set.
Complete metric space	A metric space in which every Cauchy sequence converges within the space.
Complete market	A market in which every admissible claim can be replicated.
Teacher-student model	A controlled learning setup in which a known teacher generates data and a student is trained to recover its function or predictions.
Barrier option	An option activated or terminated when the underlying crosses a specified level.
MLP	A multilayer perceptron: a feedforward composition of affine maps and nonlinear activations.
Universal approximation theorem	A family of results showing that suitable neural networks are dense in specified function classes on compact domains.

Term	Concise definition
Functional form	The chosen mathematical shape relating inputs to outputs.
Least-squares Monte Carlo	A regression-based backward algorithm for estimating continuation values and American exercise decisions.
SDE	A stochastic differential equation combining drift and random increments, interpreted in integral form.
Linear model	A model linear in its parameters or inputs, depending on the stated convention.
Nonlinear model	A model with nonlinear dependence on inputs, parameters, or both.
Signature series	The hierarchy of iterated integrals of a path, encoding ordered path information.
Reset contract	A contract whose strike, coupon, spread, or other term changes under a scheduled or event-driven rule.
Shock	An unexpected random innovation, modeled continuously by diffusion or discontinuously by jumps.
Spline	A piecewise polynomial joined with continuity constraints at knots.
Polynomial	A finite sum of powers; a global basis that is linear in its coefficients.
Jacobian stability	The degree to which the Jacobian and its smallest singular values remain well behaved under perturbations.
Convolution	A weighted aggregation of shifted values using a kernel.
Stochastic calculus	Integration and differential rules for random processes with nonzero quadratic variation.
Path-dependent	Depending on the history of a process, not only its terminal state.
Risk-neutral simulation	Simulation under a pricing measure in which discounted tradable gains are martingales.
Itô's lemma	The stochastic chain rule containing a second-derivative correction from quadratic variation.
Itô's isometry	The identity equating the second moment of an Itô integral with the time integral of the squared integrand.

Chapter B

Formula sheet

Probability and stochastic processes

$$
\begin{aligned}
\mathbb{E}[X] &= \int_\Omega X \,\mathrm{d}\mathbb{P}, \\
\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] &= \mathbb{E}[X], \\
\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] &= \mathbb{E}[X \mid \mathcal{H}], \quad \mathcal{H} \subseteq \mathcal{G}, \\
\mathbb{E}[M_t \mid \mathcal{F}_s] &= M_s, \quad s \leq t, \\
W_t - W_s &\sim \mathcal{N}(0, t-s), \\
[W]_T &= T.
\end{aligned}
$$

Itô calculus and SDEs

$$
\begin{aligned}
\mathbb{E}\left[\left(\int_0^T H_t \,\mathrm{d}W_t\right)^2\right] &= \mathbb{E}\int_0^T H_t^2 \,\mathrm{d}t, \\
\mathrm{d}X_t &= a\,\mathrm{d}t + b\,\mathrm{d}W_t, \\
\mathrm{d}f(t, X_t) &= \left(f_t + a f_x + \frac{1}{2} b^2 f_{xx}\right)\mathrm{d}t + b f_x \,\mathrm{d}W_t, \\
S_t &= S_0 \exp\left[(\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t\right], \\
X_{n+1} &= X_n + b(t_n, X_n)\Delta t + \sigma(t_n, X_n)\sqrt{\Delta t} Z_n.
\end{aligned}
$$

Option payoffs and parity

$$
\begin{aligned}
\text{Call payoff} &= (S_T - K)^+, \\
\text{Put payoff} &= (K - S_T)^+, \\
C - P &= S e^{-q\tau} - K e^{-r\tau}, \\
\text{Up-and-out call payoff} &= (S_T - K)^+ \mathbf{1}_{\{\max_{t \leq T} S_t < B\}}.
\end{aligned}
$$

Risk-neutral valuation and Black-Scholes

$$
\begin{aligned}
V_t &= B_t \mathbb{E}^{\mathbb{Q}}\left[\frac{H}{B_T}\,\middle|\,\mathcal{F}_t\right], \\
\frac{\mathrm{d}S_t}{S_t} &= (r-q)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}, \\
0 &= V_t + \tfrac{1}{2}\sigma^2 S^2 V_{SS} + (r-q)SV_S - rV, \\
C &= Se^{-q\tau}N(d_1) - Ke^{-r\tau}N(d_2), \\
d_1 &= \frac{\log(S/K) + (r - q + \tfrac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \\
d_2 &= d_1 - \sigma\sqrt{\tau}, \\
P &= Ke^{-r\tau}N(-d_2) - Se^{-q\tau}N(-d_1).
\end{aligned}
$$

Greeks

$$
\begin{aligned}
\Delta_{\text{call}} &= e^{-q\tau}N(d_1), \\
\Gamma &= \frac{e^{-q\tau}\phi(d_1)}{S\sigma\sqrt{\tau}}, \\
\nu &= Se^{-q\tau}\phi(d_1)\sqrt{\tau}.
\end{aligned}
$$

Volatility surfaces

$$
\begin{aligned}
C^{\text{BS}}(\sigma_{\text{imp}}) &= C^{\text{mkt}}, \\
\frac{\mathrm{d}S_t}{S_t} &= (r-q)\,\mathrm{d}t + \sigma_{\text{loc}}(t,S_t)\,\mathrm{d}W_t^{\mathbb{Q}}, \\
\sigma_{\text{loc}}^2(T,K) &= \frac{\partial_T C + (r-q)K\partial_K C + qC}{\tfrac{1}{2}K^2\partial_{KK}C}, \\
\partial_{KK}C(K,T) &= e^{-rT} f_{S_T}^{\mathbb{Q}}(K).
\end{aligned}
$$

Inverse problems and geometry

$$
\begin{aligned}
\mathcal{P}(\theta) &= (P_1(\theta), \dots, P_m(\theta)), \\
\delta p &\approx J(\theta)\delta\theta, \\
\ker J &= \{v : Jv = 0\}, \\
\kappa_2(J) &= s_{\max}/s_{\min}, \\
\widehat{\theta}_\lambda &= \arg\min_\theta \left\| W^{1/2}(\mathcal{P}(\theta) - p) \right\|^2 + \lambda R(\theta).
\end{aligned}
$$

Approximation and signatures

$$
\begin{aligned}
f_\beta(x) &= \sum_j \beta_j \phi_j(x), \\
f_\theta(x) &= \sum_{j=1}^{m} a_j \rho(w_j^\top x + b_j) + c, \\
(f * k)(t) &= \int f(t-u) k(u)\,\mathrm{d}u, \\
S^{i_1,\dots,i_n}(X) &= \int_{0<t_1<\cdots<t_n<T} \mathrm{d}X_{t_1}^{i_1} \cdots \mathrm{d}X_{t_n}^{i_n}, \\
\mathcal{S}(X * Y) &= \mathcal{S}(X) \otimes \mathcal{S}(Y).
\end{aligned}
$$

Chapter C

Exercises with solution sketches

C.1 Foundations

Exercise 1: A generated sigma-algebra

Let $\Omega = \{1, 2, 3, 4\}$ and $A = \{1, 2\}$. List the sigma-algebra generated by A.

Solution sketch. Closure under complements gives $A^c = \{3, 4\}$. Together with $\varnothing$ and Ω, this is already closed under countable unions:

$$\sigma(A) = \{\varnothing, A, A^c, \Omega\}.$$

Exercise 2: Borel measurability

Show that $[a, b)$ is a Borel subset of $\mathbb{R}$.

Solution sketch. Closed sets and open sets are Borel. One representation is

$$[a, b) = (-\infty, b) \cap [a, \infty),$$

a finite intersection of Borel sets. Alternatively,

$$[a, b) = \bigcap_{n=1}^{\infty} (a - 1/n, b).$$

Exercise 3: Nondimensionalization

Use homogeneity of a call payoff to show that, in any multiplicatively homogeneous model, $C(S, K) = Sc(K/S)$ when other parameters are fixed.

Solution sketch. Homogeneity means $C(\lambda S, \lambda K) = \lambda C(S, K)$. Choose $\lambda = 1/S$:

$$C(1, K/S) = C(S, K)/S.$$

Define $c(x) = C(1, x)$.

C.2 Processes and stochastic calculus

Exercise 4: Expected quadratic variation

For a uniform partition of $[0, T]$ into n intervals, compute the expectation and variance of

$$Q_n = \sum_{i=1}^{n} (\Delta W_i)^2.$$

Solution sketch. Each $\Delta W_i \sim \mathcal{N}(0, T/n)$. Hence

$$\mathbb{E}[(\Delta W_i)^2] = T/n, \qquad \operatorname{Var}((\Delta W_i)^2) = 2(T/n)^2.$$

Independence gives

$$\mathbb{E}[Q_n] = T, \qquad \operatorname{Var}(Q_n) = 2T^2/n.$$

Thus $Q_n \to T$ in L^2.

Exercise 5: An exponential martingale

Show that

$$M_t = \exp(\sigma W_t - \tfrac{1}{2}\sigma^2 t)$$

is a martingale.

Solution sketch. For $s < t$, write $W_t = W_s + (W_t - W_s)$. The increment is independent of $\mathcal{F}_s$ and Gaussian with variance $t - s$. Therefore

$$\begin{aligned} \mathbb{E}[M_t \mid \mathcal{F}_s] &= e^{\sigma W_s - \sigma^2 t/2} \mathbb{E}[e^{\sigma(W_t - W_s)}] \\ &= e^{\sigma W_s - \sigma^2 t/2} e^{\sigma^2 (t-s)/2} = M_s. \end{aligned}$$

Exercise 6: Itô isometry

Evaluate

$$\mathbb{E}\left[\left(\int_0^T (T - t)\, \mathrm{d}W_t\right)^2\right].$$

Solution sketch. Apply Itô's isometry:

$$\int_0^T (T - t)^2 \, \mathrm{d}t = \frac{T^3}{3}.$$

Exercise 7: Logarithm of GBM

Starting from $\mathrm{d}S_t = \mu S_t \, \mathrm{d}t + \sigma S_t \, \mathrm{d}W_t$, derive the SDE for $\log S_t$.

Solution sketch. Itô's lemma with $f(S) = \log S$ gives

$$\mathrm{d} \log S_t = (\mu - \tfrac{1}{2}\sigma^2) \, \mathrm{d}t + \sigma \, \mathrm{d}W_t.$$

The negative half-variance term comes from $\frac{1}{2} f''(S)(\sigma S)^2$.

Exercise 8: Strong versus weak error

Why can a discretization have weak order one and strong order one-half without contradiction?

Solution sketch. Strong error measures pathwise distance, while weak error measures only the error of expectations of test functions. Averaging cancels some pathwise errors, so weak convergence can be faster.

C.3 Contracts and pricing

Exercise 9: Moneyness

Suppose $S_0 = 100$, $r = 4\%$, $q = 1\%$, and $T = 2$. Compute the forward $F_{0,T}$ and log-moneyness $k = \log(K/F)$ for $K = 110$.

Solution sketch.

$$F = 100e^{(0.04-0.01)2} \approx 106.1837, \qquad k = \log(110/106.1837) \approx 0.03531.$$

The option is slightly above forward at-the-money under this convention.

Exercise 10: Put-call parity

A one-year European call and put share strike $K = 100$. Spot is 100, $r = 5\%$, no dividends, and the call price is 10. What put price is required by parity?

Solution sketch.

$$P = C - S + Ke^{-rT} = 10 - 100 + 100e^{-0.05} \approx 5.1229.$$

Exercise 11: Risk-neutral expectation

Under Black-Scholes, write an exact simulation formula for S_T under $\mathbb{Q}$ and an unbiased Monte Carlo estimator of a European call.

Solution sketch.

$$S_T = S_0 e^{(r-q-\sigma^2/2)T+\sigma\sqrt{T}Z}, \qquad \widehat{C} = e^{-rT}\frac{1}{N}\sum_i (S_T^{(i)} - K)^+.$$

Exercise 12: Why Vega implies unique implied volatility

Explain why positive Vega yields a unique implied volatility inside Black-Scholes price bounds.

Solution sketch. Continuity and strict monotonicity of $C^{\mathrm{BS}}(\sigma)$ mean the equation $C^{\mathrm{BS}}(\sigma) = C^{\mathrm{mkt}}$ can have at most one solution. The limiting prices as $\sigma \downarrow 0$ and $\sigma \uparrow \infty$ determine existence within the arbitrage interval.

Exercise 13: Barrier path dependence

Construct two continuous paths with the same $S_T = 120$, strike 100, and up-and-out barrier 130, but different payoffs.

Solution sketch. One path remains below 130 and ends at 120, paying 20. Another rises above 130 before returning to 120, so it is knocked out and pays zero.

C.4 Local volatility and numerics

Exercise 14: Dupire under constant volatility

Assume the call surface exactly satisfies the Black-Scholes forward equation with constant σ. Substitute that equation into Dupire's formula.

Solution sketch. The numerator becomes $\frac{1}{2}\sigma^2 K^2 C_{KK}$, so the quotient is σ^2. Thus Dupire recovers the constant input volatility wherever $C_{KK} > 0$.

Exercise 15: LSMC decision

At one exercise date, an American put has immediate payoff 8 and estimated continuation value 7.4. What does LSMC do? What if the continuation estimate is biased downward?

Solution sketch. It exercises because $8 > 7.4$. Downward bias causes excessive early exercise and can lower the value of the learned stopping policy.

Exercise 16: Monte Carlo sample size

If an estimator has standard error 0.08 with $N = 10{,}000$ independent paths, approximately how many paths are needed for standard error 0.02?

Solution sketch. Standard error scales as $N^{-1/2}$. Reducing it by a factor four requires $4^2 = 16$ times as many paths: about 160,000.

C.5 Identifiability and geometry

Exercise 17: Rank and kernel

Let

$$J = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}.$$

Find its rank and describe its kernel dimension.

Solution sketch. The second row is twice the first, so rank is one. By rank-nullity in $\mathbb{R}^3$, the kernel has dimension $3 - 1 = 2$. Only one independent parameter combination is locally observed.

Exercise 18: A gauge symmetry

For $f_{a,b}(x) = abx$, compute the Jacobian of the scalar output at fixed $x \neq 0$ and identify a local null direction.

Solution sketch.

$$J = (bx, ax).$$

A vector $v = (a, -b)$ satisfies $Jv = abx - abx = 0$. It is tangent to the gauge orbit $(a, b) \mapsto (e^t a, e^{-t} b)$.

Exercise 19: Full rank but poor conditioning

Can a square Jacobian be full rank and still make inversion unstable? Give an example.

Solution sketch. Yes. For

$$J = \operatorname{diag}(1, 10^{-8}),$$

the determinant is nonzero, but the condition number is 10^8. Noise in the second output is amplified by 10^8 when recovering the second parameter.

C.6 Approximation and paths

Exercise 20: Linear in parameters

Is $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2$ a linear or nonlinear model?

Solution sketch. It is nonlinear in x but linear in parameters β. Ordinary linear least squares applies once the features $(1, x, x^2)$ are constructed.

Exercise 21: Universal approximation

Why does the universal approximation theorem not imply that gradient descent will recover a good approximation from 20 noisy observations?

Solution sketch. The theorem concerns existence in a function class, not optimization, finite-sample generalization, noise, regularization, conditioning, or extrapolation.

Exercise 22: Signature order

For the two planar paths "right then up" and "up then right", compare level-one and the cross level-two terms.

Solution sketch. Both have level-one displacement $(1, 1)$. For right-then-up, $S^{1,2} = 1$ and $S^{2,1} = 0$; for the reverse order, these values swap. The antisymmetric part records orientation/order.

Chapter D

Experiment design and reproducibility

D.1 Environment

The figures in this guide were generated with a fixed pseudo-random seed, `20260914`, using:

- Python 3.13.5;
- NumPy 2.3.5;
- SciPy 1.17.0;
- Matplotlib 3.10.8;
- scikit-learn 1.8.0.

Small floating-point differences can occur across platforms and linear-algebra libraries, but the qualitative results and quoted precision should be reproducible.

D.2 Experiment inventory

Experiment	Main parameters	Reported result
Brownian quadratic variation	$T = 1$, up to 2,048 intervals, 4,000 trials	mean 1.00014, sd 0.03150
GBM terminal law	$S_0 = 100$, $\mu = 7\%$, $\sigma = 25\%$, $T = 1$, 120,000 samples	sample mean 107.3796, theory 107.2508
Itô isometry	$H_t = t$, $T = 1$, 60,000 paths	estimate 0.33268, exact 1/3
European-call Monte Carlo	$S_0 = K = 100$, $r = 3\%$, $\sigma = 22\%$, 400,000 paths	10.2162 ± 0.0247 s.e.; exact 10.1871
Dupire recovery	analytic BS surface, true $\sigma = 24\%$	median recovered 0.2400022
Barrier option	$B = 130$, 252 dates, 300,000 paths	price 2.9611, survival 77.43%
American put LSMC	50 dates, 110,000 paths, cubic basis	American 7.9355, European 7.4589
Price-map Jacobian	two strikes, parameters (σ, r)	singular values 81.2430, 15.4219
Function approximation	17 samples, polynomial/spline/MLP	RMSE 0.00166/0.000215/0.00833

Particle system	120 mean-field particles, final mean 0.02385, sd 0.42859 $T = 4$

D.3 Core simulation patterns

Exact GBM terminal simulation

```
z = rng.normal(size=n_paths)
ST = S0 * np.exp((drift - 0.5*sigma**2)*T
                 + sigma*np.sqrt(T)*z)
```

Euler-Maruyama

```
for n in range(n_steps):
    z = rng.normal(size=n_paths)
    x += b(t[n], x)*dt + sigma(t[n], x)*np.sqrt(dt)*z
```

Central finite-difference sensitivity

```
derivative = (price(theta + h) - price(theta - h)) / (2*h)
```

Dupire finite differences

```
CT = np.gradient(C, dT, axis=0, edge_order=2)
CK = np.gradient(C, dK, axis=1, edge_order=2)
CKK = np.gradient(CK, dK, axis=1, edge_order=2)
local_var = (CT + (r-q)*K*CK + q*C) / (0.5*K**2*CKK)
```

D.4 Reproducibility cautions

- The barrier experiment uses discrete daily monitoring. It is not a continuously monitored analytic benchmark.
- The LSMC estimate depends on basis, exercise dates, and random sample. It is intended as an explanatory experiment rather than a production valuation.
- Dupire recovery is performed on noise-free analytic prices and therefore illustrates a best-case numerical inversion.
- The neural-network fit depends on optimization settings; the comparison demonstrates functional-form bias, not a universal model ranking.
- Figures displaying stochastic paths are individual realizations and should not be interpreted as forecasts.

D.5 Suggested extensions

Re-run the experiments after changing exactly one element:

- add quote noise before Dupire inversion and compare smoothing penalties;
- compare discrete barrier monitoring with Brownian-bridge correction;
- replace the LSMC polynomial basis with splines, MLPs, or path-signature features;
- reduce the smallest Jacobian singular value by choosing two nearly redundant instruments;
- add jumps and compare vanilla calibration with barrier prices;
- use antithetic variates and measure variance reduction empirically.

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